Role of YqhY in lipid biosynthesis regulation in *Bacillus Subtilis*

Bachelor End Project

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Abstract

The regulation of membrane biosynthesis stands as a critical determinant of cellular homeostasis, orchestrating the dynamic adjustments in lipid availability vital for cellular functions. This regulation involves a complex interplay of transcriptional and post-translational factors, shaping the synthesis of membrane components at the gene expression level and influencing the activity or stability of these lipids and proteins. Varied regulatory mechanisms exist for distinct segments of membrane biosynthesis pathways, such as those governing fatty acid and phospholipid synthesis. Our study focused on unraveling the intricate dynamics of YqhY in modulating acetyl-CoA carboxylase (ACC) activity, proposing it as a feedback inhibitor influencing malonyl-CoA production. Despite challenges in pinpointing the precise regulatory mechanism, our investigations suggested an interaction between YqhY and ACC at the poles. We speculate that when operating as an inhibitor, YqhY takes on a focal form; conversely, in an inactive state, it relocates to the cytoplasm, allowing for an increase in ACC activity. Employing GFP tagging in *Escherichia coli* and attempted analysis in *Bacillus subtilis*, we partially validated prior findings, shedding light on YqhY's behavior under fatty acid synthesis blockage and its role in modulating ACC activity. These findings advance our understanding of the interplay between YqhY and ACC, opening avenues for future exploration into bacterial stress responses and regulatory mechanisms.

Key words: YqhY, fatty acid pathway, regulation, *Bacillus subtilis*, Acetyl-CoA Carboxylase

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1
Introduction

1.1. The membrane

Theodor Schwann and Matthias Jakob Schleiden, two German scientists, developed the cell theory in 1839 and defined the cell as the fundamental building block of life [1]. Since then, the field of biology has accepted this idea and now uses the cell as the base to define what is alive. The membrane is a fundamental component of cells, serving as a dynamic boundary responsible for mechanisms such as communication, transport and protection. Comprised primarily of lipids and proteins, it forms a semi-permeable barrier that regulates the passage of molecules, ions, and nutrients in and out of the cell. In most organisms, phospholipids, with hydrophilic heads and hydrophobic tails, are the primary lipid components, arranging themselves into a bilayer structure. Proteins embedded within this lipid matrix perform various functions, including transport, signaling, and structural support [2][3]. Gram-positive bacteria, including *Bacillus subtilis* (*B. subtilis*), possess an inner membrane composed of phospholipids and membrane proteins surrounded by layers of peptidoglycan, forming a robust cell wall. In contrast, Gram-negative bacteria have both an inner membrane and an outer membrane, with a thin layer of peptidoglycan sandwiched between them [Figure 1.1] [4].

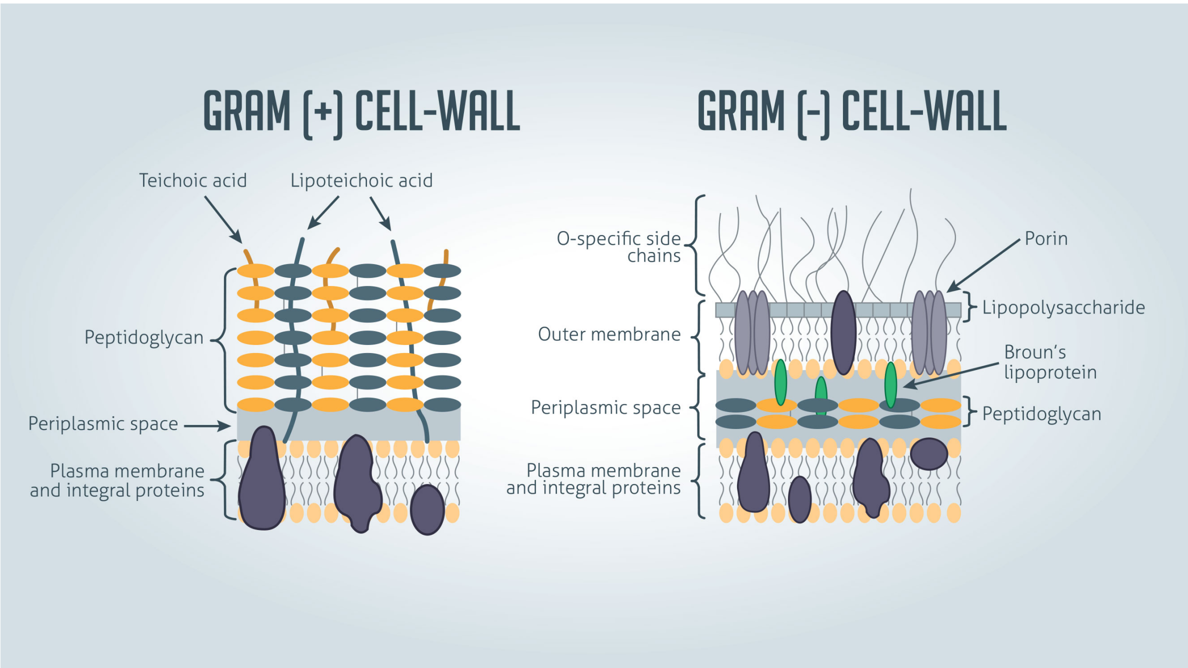

Figure 1.1: Gram-positive bacteria vs gram-negative bacteria membrane structure [4]

1.1.1 The membrane can regulate the production flow of lipids

The membrane possesses many important characteristics which allow it to function; among them adaptability is fundamental for various cellular activities. The ability of cellular membranes to adapt is essential for coordinating the complex dance of molecular fluxes that occur inside the bacterial microenvironment. In particular, the components of the membrane pathway have an extraordinary ability to dynamically modify the flow of biomolecules, serving as perceptive gatekeepers sensitive to the subtle requirements of the bacterial environment. The membrane's precisely calibrated regulatory mechanism enables it to control the synthesis of particular components, such as fatty acids and phospholipids, in response to the bacteria's constantly shifting requirements. The membrane's capacity to control molecular flow represents an intricate regulatory mechanism, guaranteeing that the bacterial environment stays precisely balanced and responsive to its needs [5].

One mechanism through which bacteria modulate the throughput of membrane components is via the regulation of lipid biosynthetic pathways. Enzymes involved in both fatty acid and phospholipid synthesis are often subject to tight regulation, allowing for precise control of lipid production in response to the cellular and environmental cues [6]. The ability of bacterial membranes to adapt is not only a survival strategy but also a means to optimize cellular function. *B. subtilis* is a great example of this mechanism since it ensures efficient energy utilization, nutrient uptake, and overall cellular homeostasis, by adjusting fatty acids and phospholipids flux according to growth rate, environmental factors, and precursor availability [7][8]. Furthermore, exploring the regulatory exchange of membrane adaptation and fatty acid synthesis in *B. subtilis* not only reveals the physiological complexities of this Gram-positive bacterium, but also gives a foundation for understanding broader bacterial regulation mechanisms. This insight is critical for determining how the intricate balance of cellular components is maintained and regulated, laying the groundwork for future advances in antibacterial research and therapy techniques.

1.2. *Bacillus subtilis* membrane synthesis

Membrane synthesis in *B. subtilis* unfolds in three distinct stages. The initial step involves the creation of fatty acids, the building blocks of lipids. During this phase, precursor molecules are enzymatically modified to form hydrophobic chains. Subsequently, these fatty acids integrate into phospholipids, which constitute a major component of cellular membranes. The final step involves the differentiation of these phospholipids into saturated and unsaturated forms. By sequentially navigating through these three stages, cells can precisely regulate the composition of their lipid membranes, tailoring them to specific physiological requirements and environmental conditions [6][9].

1.2.1 Fatty acid biosynthesis

As the first step of the membrane synthesis pathway, fatty acid production is carried out in an organised manner guided by a number of enzymes [Figure 1.2]. The process begins with the carboxylation of acetyl-CoA, which is then converted into malonyl-CoA by the enzyme acetyl-CoA carboxylase (ACC). FabD performs an essential modification on malonyl-CoA, replacing its CoA moiety with an acyl carrier protein (ACP) to make malonyl-ACP. Afterwards, malonyl-ACP undergoes an elongation cycle, led by the type II fatty acid synthase (FASII). The process involves a cascade of enzymes each contributing to the elongation of fatty acids, thus completing the fatty acid biosynthesis pathway [9].

1.2.2 Phospholipids biosynthesis

Once the fatty acids reach the required length, the phospholipids biosynthesis starts. Through this process, the fundamental components of phosphatidic acid (PA) are produced. PA is generated when two fatty acids are combined with the hydroxyl groups of glycerol-3-phosphate (G3P) during an esterification process. PlsX performs the first step in the phospholipid formation process. This enzyme converts acyl-ACP into acyl-phosphate. The following phase involves PlsY coordinating the acyl group's switch from acyl-phosphate to 1-acyl-G3P. Finally, PlsC completes the picture by using acyl-ACP to acylate the 2-position of the 1-acyl-G3P. This results in the synthesis of phosphatidic acid, which is an important building block for the synthesis of various and vital phospholipids found in biological membranes [10][11]. As the phospholipids are produced, they differentiate into saturated and unsaturated thanks to the DesR/DesK system [12].

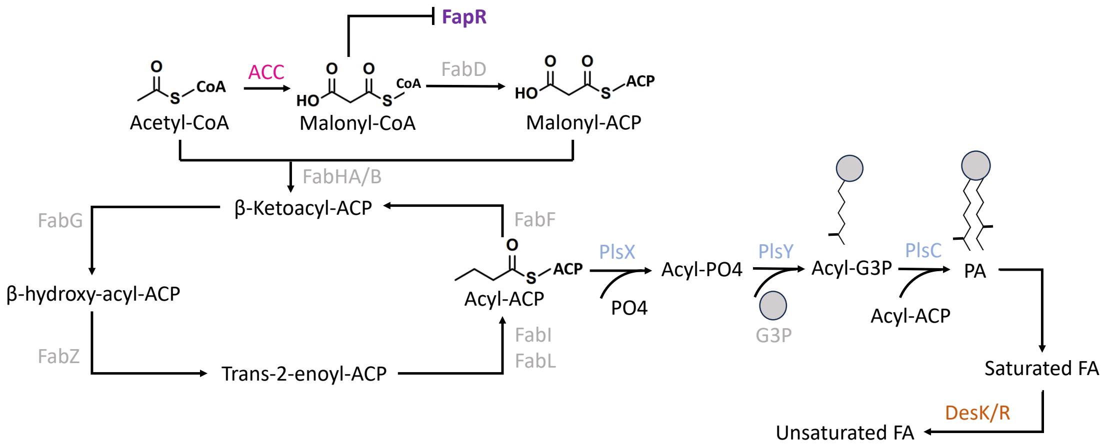

Figure 1.2: Overview of the *B. subtilis* FASII pathway and FA integration into phospholipids

1.2.3 Mechanisms governing these pathways

While the reactions involved in membrane formation, notably those regarding fatty acid and phospholipid production, are well-defined, the precise molecular mechanisms guiding their regulation remain unknown. Despite knowing that the enzymes responsible for both syntheses are precisely regulated, the exact mechanisms, signaling cascades, and key regulatory components that orchestrate these activities are not well understood. Unravelling the specifics of this complex regulatory network represents substantial challenge for bacterial physiology.

1.3. Regulation of the membrane synthesis pathway

Regulating membrane biosynthesis is a highly intricate and crucial aspect of cellular homeostasis, ensuring the precise control of molecular fluxes to meet the dynamic needs of the cell. The regulation of membrane biosynthesis involves an interplay of various factors, with transcriptional and post-translational regulators playing key roles. Transcriptional regulators influence the synthesis of membrane components at the level of gene expression, determining the quantities of proteins involved in membrane biosynthesis [13]. Post-translational regulators, on the other hand, modulate the activity or stability of these proteins after they have been synthesized [14]. An example of a post-translational regulator can be found in *E. coli*'s PlsB, an enzyme that combines the enzymatic functions typically carried out individually by PlsX and PlsY in *B. subtilis*. Interestingly, experimental data indicates that the rate at which this crucial process is synthesised *in vivo* is independent of the concentration of substrates or enzymes. This suggests that PlsB functions through an unknown, post-translational regulatory mechanism [15].

Notably, different regulatory mechanisms may exist for distinct stages of membrane biosynthesis pathways, such as those governing fatty acid and phospholipid synthesis. Multiple regulators could act in concert, balancing between saturated and unsaturated fatty acids or influencing the flux of specific phospholipids into the membrane. This regulation ensures the adaptability and responsiveness of cellular membranes to changing environmental conditions, facilitating essential cellular functions and maintaining overall cell integrity.

1.3.1 Transcriptional regulation in *Bacillus subtilis*

A key transcriptional regulator, FapR, holds pivotal role in managing fatty acid biosynthesis in *B. subtilis*. FapR acts as a global transcriptional repressor, regulating the expression of the *fap regulon*, encompassing genes involved in fatty acid and phospholipid synthesis such as *fabHA-fabF, fapR-plsX-fabD-fabG, fabI, fabHB, yhfC,* and *plsC*. The binding sites for FapR lie within the promoter regions upstream of these genes, with their interaction exclusively disrupted by malonyl-CoA. This compound, an intermediary in fatty acid synthesis, functions as a signaling molecule, allowing the cell to measure the concentration of fatty acids and modulate the expression of the *fap regulon* accordingly. This mechanism underscores malonyl-CoA's novel role in fine-tuning FapR activity, thereby regulating membrane lipid homeostasis. In the intricate dance of cellular regulation, Malonyl-CoA takes center stage, pulling the strings on the enzymes of the *fap-operon*, intriguingly exempting those responsible for its own production. This curious absence prompts a captivating question: what regulates the production of Malonyl-CoA itself [9][11][16]?

1.3.2 Acetyl-CoA Carboxylase

To comprehend how malonyl-CoA is regulated it is necessary to understand ACC's structure and functionality. ACC plays a pivotal role in fatty acid biosynthesis, operating as a biotin-dependent enzyme that regulates the irreversible conversion of acetyl-CoA into malonyl-CoA. This reaction involves a two-step catalytic process mediated by the biotin carboxylase (BC) and carboxyltransferase (CT) activities within ACC. This protein constitutes a holoenzyme that comprises four subunits arranged in two sub-complexes, AccB-AccC and AccA-AccD, all of which are conserved [Figure 1.3] [9][17].

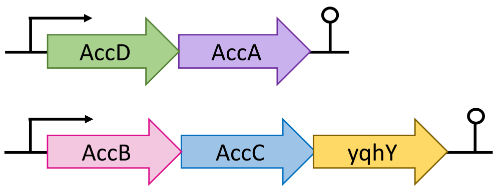

Figure 1.3: Operons of ACC sub-complexes

The carboxylation process involves an ATP-dependent homodimeric biotin carboxylase (AccC), a biotin carboxyl carrier protein (AccB) carrying the biotin cofactor, and bicarbonate as the source of CO_2. Subsequently, the carboxyl group transfer from biotin to acetyl-CoA is facilitated by an $\alpha 2\beta 2$ heterotetrameric carboxyltransferase domain (AccAD) leading to the formation of malonyl-CoA [7], with the α-subunit encoded by AccA and the β-subunit by AccD [9].

Although the exact subcellular location of the whole ACC complex is still unknown, new research has provided insight into the sublocalization of AccA. Specifically, AccA has been identified in distinct foci concentrated at the poles of the bacterial cell [Supplementary Figure A.4] [17]. Considering the formation of a heterodimer between AccA and AccD, it is conceivable that AccD might also localize at the poles, in tandem with AccA. The specific positioning of ACC could help in understanding its regulation, by offering clues about potential interactions with regulatory proteins.

1.4. YqhY: a potential regulator of fatty acid synthesis

As depicted in Figure 1.3, the ACC operons contain a fifth gene, *yqhY*, identified within the genome of *B. subtilis* as an essential gene of unknown function. This gene encodes for YqhY which is a highly conserved and expressed protein: around 22,500 molecules per cell [18]. YqhY is part of the alkaline shock protein 23 (Asp23)/Gls24 family, a group of proteins involved in the cellular response to envelope stress in certain organisms [19].

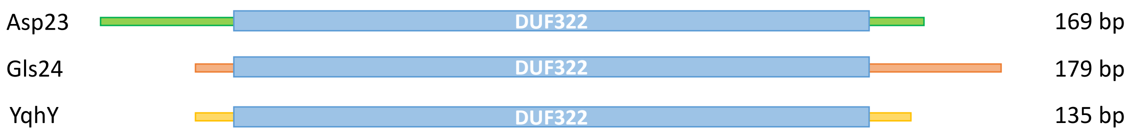

Figure 1.4: Protein alignment of Asp23 (green), Gls24 (orange) and YqhY (yellow), with visual representation of the DUF322 domain (blue).

All members of the Asp23/Gls24 family share a common structural feature, comprising a solitary domain of unknown function (DUF) 322, encompassing approximately 110 amino acids. Surrounding this core domain, there are distinctive N- or C-terminal extensions, exhibiting variations in their lengths across different family members [Figure 1.4 and Supplementary Figure A.1]. These extensions contribute to the overall diversity and functional nuances within the Asp23/Gls24 family, potentially influencing the specific roles and interactions of these proteins within biological processes. Overall, the DUF322 domain is a highly conserved entity within the realm of gram-positive bacteria [Supplementary Figure A.2]. This domain has been intricately linked with numerous essential proteins, adding an extra layer of significance to its presence [20]. The compelling question arises: could the DUF322 family have a role in regulating the membrane synthesis? Determining possible similarities among the proteins that contain the DUF322 domain is essential to comprehending its intricate functions. This effort provides light on the complex mechanisms that underlie gram-positive bacteria and creates opportunities for a deeper understanding of the basic biology that underpins this fascinating field.

In particular, Asp23, a protein abundantly present in *Staphylococcus aureus*, holds a prominent position in the cellular landscape. Despite lacking recognizable transmembrane spanning domains, Asp23 localizes in close proximity to the cell membrane except the sites of septum formation. The absence of transmembrane domain, led to the identification of AmaP (Asp23 Membrane Anchoring Protein), another protein of the DUF322 family, as the membrane anchor for Asp23 [Figure 1.5.A]. In fact, there is a direct protein-protein interaction between the entirety of AmaP and the DUF322 domain and the C-terminal of Asp23. Furthermore, deletion of Asp23 and its delocalization caused by AmaP deletion [Figure 1.5.B] resulted in an upregulation of the cell wall stress response [20]. Lastly, Asp23 has been identified as a protein that accumulates in the soluble protein fraction of *S. aureus* when exposed to alkaline shock [21].

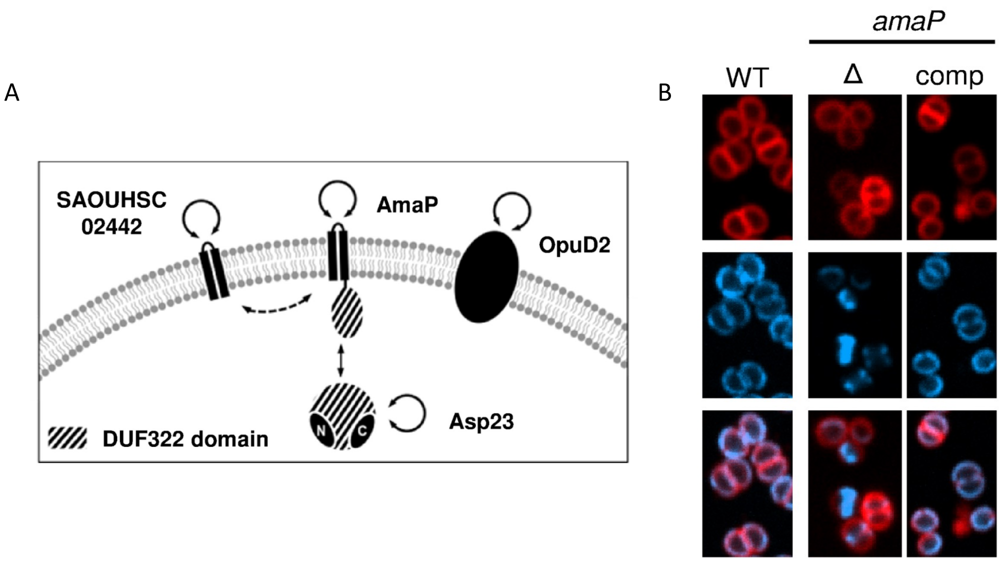

Figure 1.5: A. Interaction between Asp23 and AmaP at the membrane. B. Effects of AmaP deletion (Δ), compared to the wildtype (WT). In parallel also the complement (comp). In red the membrane, in blue Asp23.

Experimental observations have revealed that *in vitro*, Asp23 demonstrates the ability to form filaments, while *in vivo*, oligomerization has been identified. The interaction between two Asp23 molecules might occur through the engagement of their respective DUF322 domains [22]. This hypothesis gains support from Asp23's structural composition, consisting of only an N-terminal, a DUF322 domain, and a C-terminal. Moreover, the potential for interaction between DUF322 domains is substantiated by the interaction between AmaP and Asp23, where two DUF322 proteins were shown to engage in a molecular interaction [20].

YqhY, similar to Asp23, contains the DUF322 domain [Figure 1.4], which adds an intriguing layer to its potential functionality, raising the possibility of various protein-protein interactions. One plausible scenario is the prospect of self-interaction, akin to the filament formation observed in the case of Asp23 [22]. This would imply that YqhY has the potential to create filamentous structures, contributing to its cellular role. Furthermore, upon conducting a TMHMM (Trans-Membrane Helix prediction using Hidden Markov Models) analysis, it became apparent that YqhY as well does not possess a transmembrane domain [23]. Considering the precedent set by the interaction between AmaP and Asp23 within the DUF322 family [20], it's conceivable that YqhY might engage in interactions with another protein of the same family which influences its localization. Given these structural similarities and its membership to the Asp23/Gls24 family, it is conceivable that YqhY shares an analogous potential mechanism in the regulation of membrane homeostasis, paralleling the characterized structure of Asp23.

1.4.1 Role of YqhY in membrane regulation

It has been shown that YqhY concentrates in foci predominantly near the poles of bacteria [Figure 1.6.A]. Additionally, its involvement in the coordination of the ACC complex and consequent impact on lipid biosynthesis is supported by its inclusion in the accC-accB-yqhY operon [Figure 1.3]. Interestingly, the loss of YqhY validates this theory. Firstly, there was a rapid acquisition of suppressor mutations affecting all ACC subunits in their functional domains [Figure 1.6.B]. The observed suppressor mutations in these domains of ACC subunits suggest a compensatory mechanism initiated by the cells to counteract the deleterious effects caused by the loss of YqhY. Moreover, there was a formation of lipophilic clusters in the polar regions of the cells indicating an increase in lipid production, and therefore a higher activity of ACC [Figure 1.6.C]. Finally, this hyperactivity of the enzyme resulted in depletion of the cellular acetyl-CoA pool [Supplementary Figure A.3] [7].

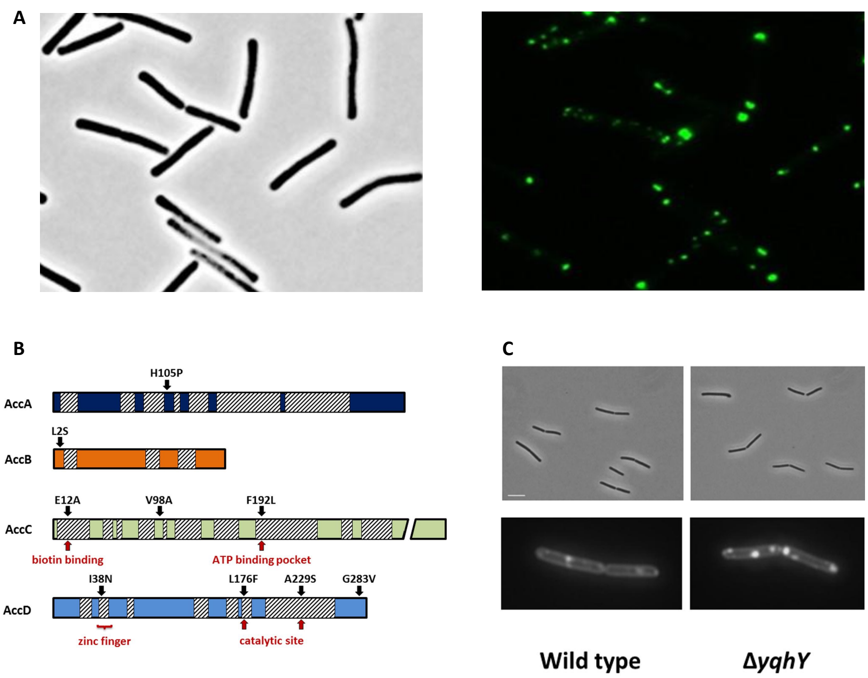

Figure 1.6: Overview of the results obtained by Tödter et al. A. Localization of YqhY. The protein was fused to a monomeric GFP at the C-terminal and visualized via fluorescence microscopy. It is visible that the YqhY collects in foci at the poles of the bacteria. B. Suppressor mutations in the ACC subunits caused by YqhY knockout. Functional domains depicted with diagonal lines. C. Lipophilic clusters when YqhY is lost [7].

The observed upregulation resulting from the loss of YqhY parallels the behavior exhibited by the deletion and delocalization of Asp23, as mentioned earlier. In fact, as deletion and delocalization of Asp23 causes an increase in stress gene synthesis, a deletion of YqhY causes ACC hyperactivity. Furthermore, when considering the subcellular localization, YqhY behaves similarly to Asp23, suggesting potential functional similarities among these proteins. It is known that Asp23 localizes thanks to the direct protein-protein interaction with AmaP [20]; in a similar way YqhY could localize at the poles. Additionally, considering the polar localization of AccA, there is the possibility that YqhY and the enzyme subunit may co-localize, potentially leading to a direct interaction between the two proteins. Unfortunately, the exact nature of their interaction and co-localization remains to be confirmed.

Finally, considering the structural similarity between YqhY and Asp23, it is hypothesised that YqhY may also be able to produce filaments by employing its DUF322 domain as a possible binding domain. If YqhY were to exhibit filamentation, it could introduce a novel potential regulatory mechanism. The possibility is raised that YqhY might employ filamentation as a means to regulate or inhibit the activity of ACC. This proposed regulatory role through filamentation could provide additional insights into the intricate mechanisms governing cellular processes, particularly those involving fatty acid biosynthesis. The potential inhibition of ACC by YqhY filamentation suggests a new level of control within these pathways. To comprehensively understand and validate this hypothesis, further experimental investigations would be essential.

Overall, YqhY's impact on lipid biosynthesis and ACC coordination is supported by several factors. Firstly, its inclusion in the *accC-accB-yqhY operon* [Figure 1.3], suggests that it has a direct influence on ACC. This has been confirmed in previous studies that showcased an upregulation of the enzyme in the absence of YqhY. Moreover, its resemblance to Asp23 in various aspects indicates a shared functional role, possibly involving lipid metabolism or related pathways. Furthermore, the presence of YqhY foci at the poles is associated with an involvement in lipid regulation due to the spatial correlation between these foci and sites crucial for lipid membrane synthesis and cell division. In fact, the cell poles are active regions involved in membrane bio-genesis and replication [24].

Adding depth to this observation is the noteworthy parallel observed in other bacteria, where proteins presumed to play pivotal roles in lipid regulation also exhibit a focal localization pattern. For instance, *E. coli*'s PlsB, the first enzyme in phospholipid biosynthesis, considered to be the exclusive post-translational regulator of membrane synthesis [15]. According to Greg Bokinsky's Lab the focal form of PlsB is thought to be inactive, contrasting with its active state when existing in the cytoplasm as a monomer or dimer. This intriguing similarity in the subcellular localization dynamics between YqhY and other lipid-regulating proteins across diverse bacterial species accentuates the potential universality of such regulatory mechanisms in bacterial lipid metabolism.

In a broader context, YqhY emerges as a compelling candidate for post-translationally controlling ACC, and therefore malonyl-CoA synthesis. However, despite all the previously mentioned parallels, the mechanisms that govern the interaction between YqhY, membrane dynamics, and the cellular environment remain unknown. Understanding the regulatory intricacies and molecular interplay involving YqhY, particularly in the context of membrane dynamics and cellular environment, holds the key to unraveling the finer details of lipid metabolism and its significance for the overall physiological well-being of the organism.

1.5. Research hypothesis

This study aims to elucidate the exact role played by YqhY in modulating ACC activity to maintain a balance between fatty acid synthesis and other cellular requirements. Our working hypothesis proposes YqhY as an inhibitor of ACC, responsive to malonyl-CoA concentration, thereby influencing malonyl-CoA production. However, the specific mechanism remains hard to establish. We suggest a direct interaction between YqhY and ACC at the poles, as both exhibit tendencies to form foci in this cellular region [7][17]. When operating as an inhibitor, YqhY takes on a focal form; conversely, in an inactive state, it relocates to the cytoplasm, allowing for an increase in ACC activity [Figure 1.7].

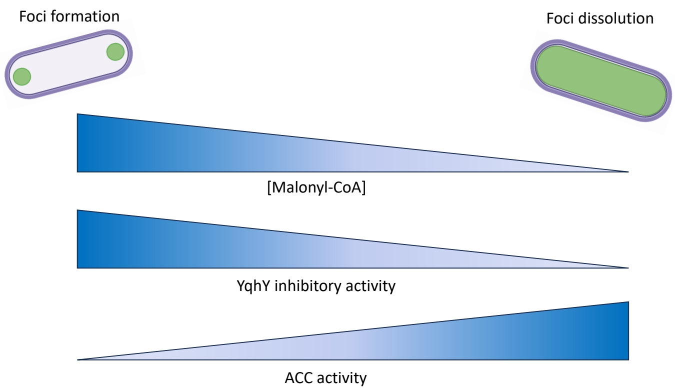

Figure 1.7: Overview of our hypothesis. When YqhY is in its focal form it is conducting its inhibitory activity, the concentration of malonyl-CoA is high and the activity of ACC is reduced. Conversely, when YqhY is in the cytoplasm, its activity is reduced, the concentration of malonyl-CoA is decreased and therefore ACC activity is increased.

This parallels the behavior observed with Asp23, which accumulates in the soluble protein fraction following exposure to alkaline shock allowing for an increase in transcription of stress response genes. Remarkably, this observed behavior in YqhY shares similarities also with PlsB in *E. coli*. When PlsB localizes at the poles, it inhibits itself through direct interaction, forming filaments; however, in the cytoplasm, it upregulates its activity, promoting increased phospholipid production. This parallel underscores the intriguing interplay between protein localization and regulatory activity within the intricate networks of bacterial lipid metabolism. Through these investigations, we aim to unravel the intricate dynamics governing YqhY and its impact on ACC activity.

1.5.1 Goals of this project

To test the aforementioned hypothesis, the following steps will be taken employing Green Fluorescent Protein (GFP) tagging:

1. YqhY will be tracked within *E. coli* to validate previous localization results, primarily observing YqhY expression as foci at the poles.
2. *E. coli* expressing YqhY will be subjected to stress-inducing scenarios including antibiotic exposure, and YqhY overexpression, to see how these affect localization.
3. YqhY will be tagged and monitored in *B. subtilis*.

These conditions will shed light on the dynamic responses of YqhY, unraveling its behavior under stress and its role in modulating ACC activity.

1.5.2 Relevance of the research

Knowing how lipid biosynthesis regulation works in *B. subtilis* is useful for several reasons:

1. **Understanding cellular homeostasis**: Studying lipid biosynthesis regulation helps us comprehend how cells maintain their overall balance, ensuring proper growth and survival. This knowledge can provide insights into the complex interactions between different cellular processes and the factors that influence them [25].
2. **Exploring post-translational regulation**: Research on *B. subtilis* has revealed the role of malonyl-CoA as a global regulator of lipid homeostasis, which induces the expression of several genes involved in lipid synthesis [25]. Understanding the post-translational mechanisms that control membrane lipid synthesis in bacteria can contribute to the development of new strategies for manipulating bacterial growth and behavior.
3. **Developing of new antibiotics**: Studying the regulation of lipid biosynthesis is valuable for the development of antibiotics. Several research articles and reviews highlight the significance of targeting lipid biosynthesis and its regulatory enzymes for the discovery of new antibacterial agents. The enzymes involved in fatty acids and phospholipid biosynthesis are considered ideal

targets for the development of new antibiotics [26][27]. In particular, the bacterial fatty acid synthesis pathway has been identified as a promising target for novel antibacterials, and various enzymes involved in this pathway have been proposed as potential targets for antibiotic development [28]. Understanding the regulation of lipid biosynthesis in *B. subtilis* can, therefore, provide essential insights for the identification and development of new antibiotic agents, contributing to the ongoing efforts to combat antibiotic resistance and infectious diseases [29].

4. **Investigating membrane fluidity**: The regulation of lipid biosynthesis can lead to the understanding of the mechanism of membrane fluidity optimization, which involves the isothermal control of the acyl-lipid desaturase enzyme [30]. This knowledge can contribute to the development of new strategies for controlling membrane fluidity in various organisms, including bacteria and eukaryotes.

Overall insight into these aspects can potentially lead to advancements in both basic and applied research in the fields of biology, biotechnology, and microbiology.

2 Materials and Methods

2.1. Strains

Various strains have been used in this study (Table A.1). Two main *E. coli* strains were used; NCM 3722 and DH5α. The NCM 3722 strain is a laboratory wild-type and a robust strain, which is ideal for expression experiments [31]. The DH5α strain is ideal for the cloning of plasmids [32]. Furthermore, *B. subtilis* 168 was used due to its high transformation efficiency [33].

2.2. Cloning of pBbE5k_yqhY_GFP2 in DH5α

For cloning experiments, Gibson Assembly method was employed, a widely recognized and efficient technique for assembling DNA fragments. The method involves the use of T5 exonuclease, a DNA polymerase, and DNA ligase in a single reaction. The success of Gibson Assembly hinges on the presence of overlapping regions at the ends of these fragments, achieved through meticulous primer design. These overlapping sequences, typically 20 to 40 base pairs in length, serve as homologous regions for efficient recombination during the assembly reaction.

2.2.1 Design of DNA constructs

Fragment preparation is a crucial step in the Gibson Assembly method. The process begins with the careful design of primers, which are then utilized in polymerase chain reaction (PCR) to amplify the plasmid backbone and desired insert(s). The PCR-amplified fragments undergo purification to eliminate any remnants of primers, nucleotides, or enzymes, ensuring a clean starting material, using first Dpn1 digestion, and subsequently Wizard® SV Gel and PCR Clean-Up System. Subsequent quantification and quality control steps verify the concentration and integrity of the fragments using both Nanodrop and Qubit.

For the experiment, BbE5k_PlsB_msGFP2 plasmid [Supplementary Figure A.5] was used as backbone to create pBbE5k_yqhY_GFP2. The PlsB sequence was removed, and *yqhY* sequence from *B. subtilis* 168 chromosome was inserted in the construct using Gibson Assembly. Polymerase chain reaction (PCR) was used to obtain most of the desired gene constructs. The specifications of the used plasmid can be found in Figure 2.1.

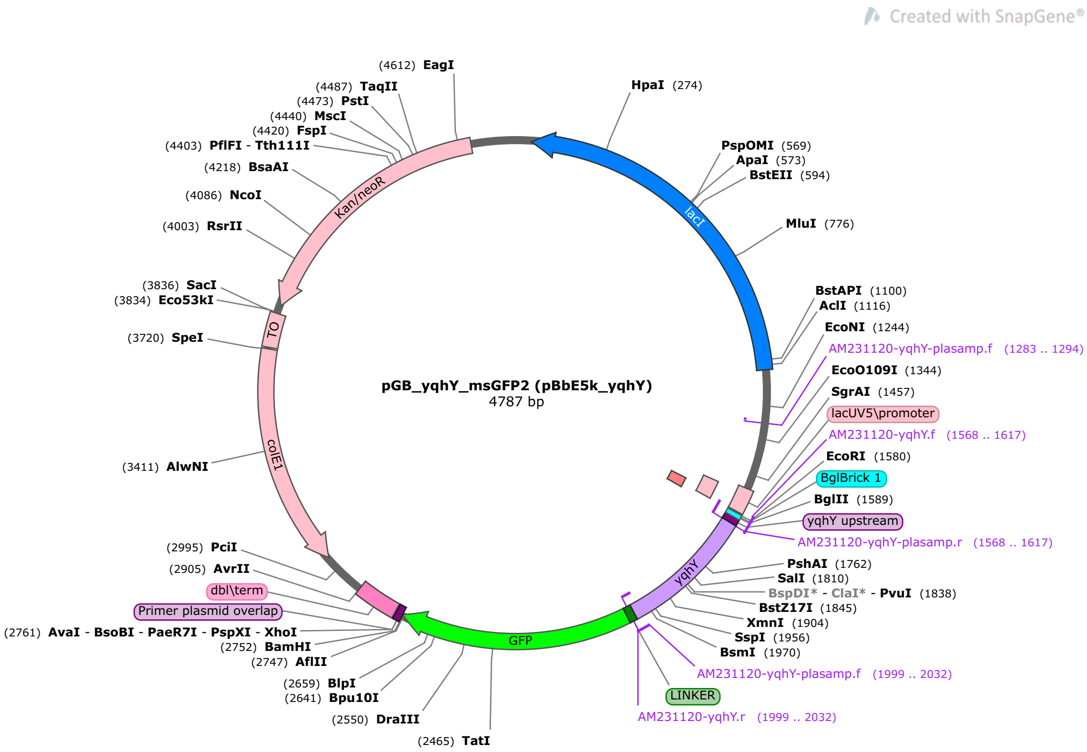

Figure 2.1: pBbE5k_yqhY_GFP2 map created with *SnapGene*. It presents an origin (colE1), resistance to kanamycin (Kan/neoR) and an IPTG inducible site (lacI).

For Gibson Assembly, two sets of primers are needed; one set amplifies yqhY and the other the vector backbone. Starting from the 5' end of the insert, a segment of about 20 nucleotides (nt) was chosen for the forward primer. Subsequently, ∼20 nt 5' overhangs complementary to the vector were added. A balanced GC concentration and melting temperature (Tm) in primer design promotes excellent annealing, which is necessary for stable primer binding throughout the assembly process [34]. For this reason, the primer was adjusted to reach a Tm of ∼58 °C [Figure 2.2]. The same process was replicated for the reverse primer. Lastly, the reverse complements of the insert primers were selected in order to design the backbone primers.

Forward Primer

5' – GTCCAAATCGACGAAGAAATGAGCGGTGGCGGTG – 3'

5' – AAAAGCCCAAGAAGTCCAAATCGACGAAGAAATGAGCGGTGGCGGTGGCAGTAACGATGA – 3'
5' – TTTTCGGGTTCTTCAGGTTTAGCTGCTTCTTTACTCGCCACCGCCACCGTCATTGCTACT – 3'

Figure 2.2: Example of forward primer designed for Gibson Assembly. Orange highlight: plasmid overhangs. Green highlight: section to amplify the insert.

For PCR amplification of the insert forward primer AM231120-yqhY.f and reverse primer AM231120-yqhY.r were used. For PCR amplification of the vector forward primer AM231120-yqhY-plasamp.f and reverse primer AM231120-yqhY-plasamp.r were used [Table 2.1].

Table 2.1: Primers used for Gibson Assembly.

Primer	Oligonucleotide sequence 5'- 3'
AM231120-yqhY-plasamp.f	GTCCAAATCGACGAAGAAATG agcggtggcggtg
AM231120-yqhY-plasamp.r	cattcaattcacctccgtaaaaa AGATCTTTTGAATTCTGAAATTGTTAT
AM231120-yqhY.f	ATAACAATTTCAGAATTCAAAAGATCT tttttacggaggtgaattgaatg
AM231120-yqhY.r	caccgccaccgct CATTTCTTCGTCGATTTGGAC

2.2.2 Phusion PCR

For each PCR reaction, a primer mix was prepared using 1 μL of 100 μM forward primer, 1 μL of 100 μM reverse primer, and 8 μL of Milli-Q (total volume: 10 μL). To ensure proper mixing the solution was pipetted up and down. ThermoFisher Scientific™ Phusion™ High–Fidelity DNA Polymerase (ThermoFisher Scientific, #F530S) and 5X Phusion™ HF Buffer were used. A standard PCR reaction, consisting of 10 μL 5X HF buffer, 1 μL 10 *nM* dNTP, 0.5 μL primer mix (0.1 μM final primer concentration), 50 *ng* template DNA (2 μL for *Bacillus subtilis* 168 genomic DNA and 0.5 μL for pBbE5k_PlsB_msGFP2 plasmid DNA), 0.5 μL Phusion polymerase and Milli-Q to have a total volume of 50 μL. To maximise the PCR reaction yield, the polymerase was added last, right before the sample area reached the right temperature, ensuring a hot start for the PCR. PCR was performed using Veriti™ 96-Well Fast Thermal Cycler (#4375305) according to a specific program [Table 2.2]. The elongation time was set according to the length of the plasmid; 30 seconds for each kbp.

Table 2.2: PCR program amplification pBbE5k_yqhY_GFP2 with primers with melting temperature 58 °C.

Stage	Stage 1 (x1)	Stage 2 (x25)			Stage 3 (x1)	
Temperature	98 °C	98 °C	58 °C	72 °C	72 °C	4 °C
Time (min)	1.00	0.15	0.30	2.15	7.00	∞
Step	Activation	Denaturation	Annealing	Extension	Final Extend	Collection

PCR cleanup was performed on all products to eliminate the vector to prevent from carrying it over into the transformation step. To do this 0.5 μL of Dpn1 was added to each PCR reaction, and then incubated at 37 °C for 30 min-1 hour. To check if the product was amplified correctly a DNA agarose gel was used. For gel analysis a gel was made with 0.5 *g* agarose, 50 *mL* of 1x TAE buffer and 5 μL Sybersafe® with TAE as running buffer. Respectively, 8 μL DNA BenchTop Ladder (Promega, #G7541) and 50 μL PCR product with 10 μL 6x Purple loading dye (NEB, #B7025) were added to the wells and the analysis run at 120 *V* for 20 minutes. If the gel showed successful results, DNA fragments were purified with Wizard® SV Gel and PCR Clean-Up System from Promega (Promega, #A9282) according to the spin-based protocol using Eppendorf Centrifuge 5425 (Eppendorf, #540500071).

2.2.3 Nanodrop

To increase the DNA concentration Eppendorf Concentrator plus™ (Eppendorf, #5305000509) was used in vacuum aqueous mode for 20 minutes. The nanodrop (NanoDrop™ 2000/2000c Spectrophotometers from ThermoFisher Scientific, #ND-2000) was then used to measure the concentration of the samples. When using the nanodrop first blank with 2 μL water, then clean the pedestal, and run a first DNA measurement with 2 μL water to confirm the zero measurement. Wipe out the water and then start measuring the DNA samples keeping track of the DNA concentration, the $A_{260/280}$ and the $A_{260/230}$. When finished, measure 2 μL water to clean the machine. Wipe the pedestal clean and close it.

2.2.4 Gibson Assembly

For Gibson Assembly reaction a reaction mix was prepared with 100 ng vector, 3x molar quantity of insert, 5 μL Gibson Assembly master mix (NEB, #E2611L) and Milli-Q to have a final volume of 10 μL. This mixture was incubated in the thermocycler for 1 hour at 50 °C.

To calculate the 3x molar quantity of insert use the following formula: $ng_{insert} = \frac{bp_{insert}}{bp_{vector}} * 3 * ng_{vector}$

Since the insert is 425 bp long, and the vector 4362 bp, the required amount of insert is 30 ng.

2.2.5 Heat shock Transformation in DH5α

Heat shock transformation was performed using *E. coli* DH5α competent cells. Aliquots of DH5α were thawed, and subsequently 10 μL of completed ligation reactions, 20 μL of ice-cold 5x KCM buffer (0.5 M KCl, 0.15 M $CaCl_2$, 0.25 M $MgCl_2$), and 70 μL ice-cold sterile water were added to them. The solution was then mixed gently and left on ice for 15-30 minutes. Using a heat block (Labnet's AccuBlock™ Digital Dry Baths) the sample was then heated to 41-43 °C for 45-60s. Next, the tubes were transferred to ice and incubated for 5 minutes. The reactions were then placed in the shaking platform (Eppendorf MixMate®, #5353000510) at 37 °C for 45 minutes since kanamycin [50 mg/mL] (Kan50) was selected as antibiotic resistance. 100 μL of the cells were then plated on Kan50 agar plates using sterile beads. Finally, the plates were left to dry, inverted and incubated at 37 °C overnight.

2.2.6 Colony PCR

To check whether the colony had the insert or not, colony PCR was performed. For colony PCR, a ThermoFisher Scientific DreamTaq Green PCR Master Mix (2X) mastermix was used (ThermoFisher, #K1082). This is a ready-to-use solution containing DreamTaq DNA Polymerase, optimized DreamTaq Green buffer, $MgCl_2$, and dNTPs. A pipette tip was used to resuspend clones from the transformation plates in 10 μL of Milli-Q. Two sequencing primers were used starting from the lacUV5/promoter and ending at the dbl/term, respectively GB030714-lacUV5.fseq1 and GB070917-dblterm.rseq1 [Table 2.3].

Table 2.3: Primers used for Colony PCR

Primer	Oligonucleotide sequence 5'- 3'
GB030714-lacUV5.fseq1	GCGTAGAGGATCGAGATCG
GB070917-dblterm.rseq1	GTGTGCCTCTAGTAGAGAGCG

Per reaction 10 μL of the 2X mastermix, 0.5 μL of forward primer (10 μM), 0.5 μL of reverse primer (10 μM), 1 μL of resuspended colony, and 8 μL Milli-Q were assembled in a total volume of 20 μL. A control group was used to check the results. The PCR was run using a thermocycler according to the program displayed in Table 2.10. Specifically, the elongation time was set according to the length of the plasmid; 1 minute for each kbp.

Table 2.4: Colony PCR program amplification of the Gibson Assembly's colonies with melting temperature 58 °C.

Stage	Stage 1 (x1)	Stage 2 (x35)			Stage 3 (x1)	
Temperature	95 °C	95 °C	50 °C	72 °C	72 °C	4 °C
Time (min)	18.00	0.30	0.30	1.15	5.00	∞
Step	Activation	Denaturation	Annealing	Extension	Final Extend	Collection

After the PCR run, the reactions were visualized using gel electrophoresis. The same gel composition as the Phusion PCR was used, see Section 2.2.2. All micro-wells were filed with 15 μL of sample and 4 μL of DNA BenchTop Ladder (Promega, #G7541). The green DreamTaq mastermix already contains the loading dye.

2.2.7 Inoculation

For the inoculation, 4 mL of Luria-Bertani (LB) media were added to a tube. Then, 4 μL of the antibiotic needed for selection, in this case Kan50 were added. A single colony was picked using either a sterile pipette tip, a sterile toothpick or a sterilized inoculation loop. The tip, toothpick or inoculation loop containing the bacterial colony was shaken in the LB media. The tubes were incubated overnight in an incubator (Eppendorf New Brunswick™ Innova® 2300/2350 Shakers, #M1250-9920) at 37 °C shaking at 180 rpm.

2.2.8 Miniprep

The plasmids were extracted from the *E. coli* cells by using Wizard® Plus SV Minipreps DNA Purification System from Promega (Promega, #A1460). The recommended protocol was followed. Measured DNA concentration using Nanodrop, see Section 2.2.3.

2.2.9 Sequencing

Finally, the plasmids were checked with sanger sequencing. When using Macrogen, only parts of the plasmid are sequenced. In this case, only the insert was sequenced using the GB030714-lacUV5.fseq1 primer. Using 1.5 mL Eppendorf Tube, added 400 ng of miniprep product (> 50 ng/μL) to 2.5 μL of lacUV5/promoter primer (10 μM). Labelled the tubes using the specific labels provided by Macrogen.

2.2.10 Heat shock Transformation in NCM 3722

Heat shock transformation was performed using *E. coli* NCM 3722 competent cells. NCM 3722 aliquots of 100 μL were thawed, and subsequently 1 μL of successfully sequenced miniprep reactions, 20 μL of ice-cold 5x KCM buffer (0.5 M KCl, 0.15 M $CaCl_2$, 0.25 M $MgCl_2$), and Milli-Q up to 200 μL were added to them. The solution was then mixed gently and left on ice for 15-30 minutes. Using a heat block the sample was then heated to 41-43 °C for 90s. Next, the tubes were transferred to ice and incubated for 3-5 minutes. The reactions were then placed in the shaking platform at 37 °C for 45-60 minutes since Kan50 (50 mg/mL) was selected as antibiotic resistance. The entire volume was then plated on Kan50 agar plates using sterile beads. Finally, the plates were left to dry, inverted and incubated at 37 °C overnight.

2.2.11 Storage in glycerol stocks

The strains were stored in -80 °C glycerol stocks. After inoculation of all strains [see section 2.2], the liquid cultures were grown to an O.D. between 0.3 and 0.4 at 37 °C with shaking (180 rpm). 1 mL of culture was then transferred to a cryotube containing 1 mL sterile 50% glycerol and stored at -80 °C.

2.3. Microscopy using *E. coli* NCM 3722

2.3.1 Microscope Settings

To analyze YqhY in *E. coli* cells, widefield fluorescence microscopy was used. An Olympus IX81F-ZDC2 microscope was used for the images with a 100x oil objective. The microscope had the following overall settings; a temperature of 37 °C, an electron multiplying gain (EM gain) of 20. For the fluorescent images a wavelength of 450 nm was used, the intensity was 50% and the exposure 400 milliseconds (ms).

2.3.2 Localization of YqhY

The strain must be grown in liquid culture in order to image the expression of YqhY. 4 μL of Kan50 and 4 mL of LB media were added to a sterile colony tube and mixed. Using a pipette tip, a colony was streaked from the desired plate and put into the tube. Until the liquid culture tube attained an optical density (OD) of 0.3–0.4, it was incubated at 37 °C and 180 rpm.

2.3.3 Preparation of cover slips

For the investigation under the fluorescent microscope, agarose cover-slips were utilised. The content of the pads was 1 μL Kan50, 500 μL of MOPS medium (0.4% glucose and 0.1% CAA), and 500 μL 2x agar (1.5%). To make a sandwich, 300 mL of agar were pipetted on top of one cover-slip (epredia, 15 x 15 mm), and then another cover-slip was put on top of the agar. The cover-slip were left to dry for roughly twenty minutes. One of the cover-slips was then taken off, and 1 μL of the (induced) cultures was pipetted on top of the agar. To keep the cover-slip from drying out, it was positioned with the agar side facing the microscope glass slide (VMR, 22 x 50 mm) and the edges sealed with clear nail polish [Figure 2.5.3].

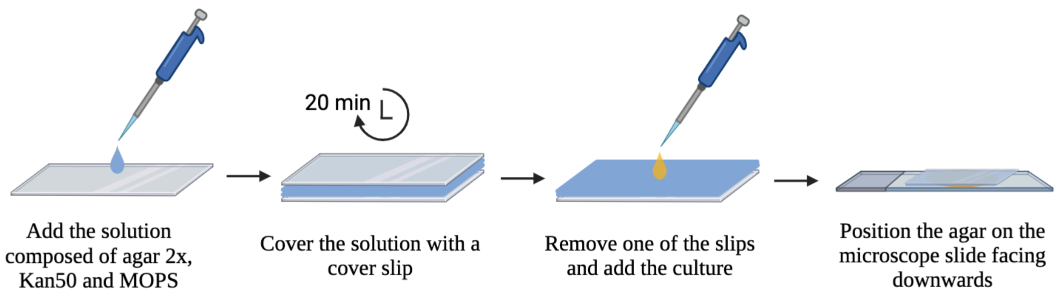

Figure 2.3: Procedure to prepare the cover slips

2.3.4 Overexpression of YqhY

A colony was inoculated using the inoculation protocol, see section 2.2.7, until an OD of 0.3-0.4 was reached. Subsequently, the culture was diluted to reach an O.D. of 0.1 and 3 μL of IPTG (100 mM) were added to induce the culture. The new culture was incubated for 30 minutes at 37 °C (180 rpm).

2.3.5 Growth rate measurements

To measure how the growth rate changes according to YqhY expression, four flasks were prepared with 30 mL of LB medium and 30 μL Kan50. For each of the four flasks, a colony was streaked from the plate and placed inside. All flasks were incubated in a water bath (Ols Aqua Pro Shaking Water Baths, Grant) at 37 °C. After 1 hour the first O.D. measurement was taken. Once an O.D. of 0.15 was reached, two of the flasks were induced using IPTG. Every 30 minutes the O.D. was measured.

2.3.6 Exposure to antibiotics

Bacteria grown with Triclosan and Cerulenin

A colony was inoculated using the inoculation protocol, see section 2.2.7, until an OD of 0.3-0.4 was reached. Subsequently, the culture was diluted to reach an O.D. of 0.15 and the antibiotic was added to the culture. One culture was prepared adding 0.75 μL of triclosan (4.4. mM), and the other with 5 μL of cerulenin (200 mg/mL). The new cultures were incubated for 30 minutes at 37 °C (180 rpm). Cover slips were prepared according to the antibiotics used. For the triclosan culture, 1.5 μL were added to the usual mixture, while for the cerulenin 9 μL, see section 2.5.3.

Bacteria exposed to Triclosan via contact with agar pads

A colony was inoculated using the inoculation protocol, see section 2.2.7, until an OD of 0.3-0.4 was reached. Cover slips were prepared according to the antibiotics used. For the triclosan pad, 1.5 μL were added to the usual mixture, see section 2.5.3.

Exposure to Mupirocin and Chloramphenicol via contact with agar pads

A colony was inoculated using the inoculation protocol, see section 2.2.7, until an OD of 0.3-0.4 was reached. Cover slips were prepared according to the antibiotics used. For both the mupirocin (15 mg/ml) and chloramphenicol (35 mg/ml) pads, 1 μL were added to the usual mixture, see section 2.5.3.

2.3.7 Analysis of microscopy images

Two approaches were taken to analyse the strains' fluorescence phenotype. Initially, a visual inspection was conducted, wherein the distinctive feature of every cell was noted and juxtaposed with the control. However since this approach can be arbitrary, another one was used.

Each cell's fluorescent spots were counted using a custom Matlab method [Supplementary Figure A.6-A.8]. The main focus of this code lies in identifying and characterizing foci within the bacterial cells seen in the fluorescence images. It employs a series of image processing steps to accomplish this. Initially, it subtracts the background fluorescence to isolate specific signals from the cells. Using the cell segmentation data obtained from MicrobeJ [35], it defines the boundaries of individual cells in the images. The analysis then identifies potential foci within each cell by applying thresholding and binarization techniques. Thresholding helps in identifying areas of interest based on their fluorescence intensity, while binarization converts the images into a binary form, highlighting these regions. Once the foci are identified, the code quantifies various parameters related to these foci within each cell. It calculates the number of foci per cell, measures the total intensity of the foci, determines the number of pixels occupied by the foci, and possibly extracts geometric properties like size and shape. There's an adaptive feature in the code that dynamically adjusts the thresholding criteria if no foci are initially detected. This adjustment ensures that even weaker foci, initially missed by the threshold, are accounted for in the analysis.

An additional segment was introduced into the code, which aims to determine the location of foci within bacterial cells. The script checks if any foci are present in the cell. If foci are detected, the centroid of the identified foci is calculated using the region properties of the binary image representing the foci. Subsequently, the centroid of the image is estimated using the region properties of the binary image representing the cell. Next, the radius of the cell is calculated based on its length. A threshold is defined for distinguishing between the "Pole" and "Septal Area" of the cell. The distance between the centroid of the foci and the centroid of the cell is then calculated using Euclidean distance. Based on this distance, the cell radius and the predefined threshold, the script categorizes the foci location as either in the "Pole" or "Septal Area" [Figure 2.4]. If no foci are present, the location is labeled as 'None'. Overall, this section contributes to the spatial characterization of foci within bacterial cells, aiding in the analysis of their distribution and localization.

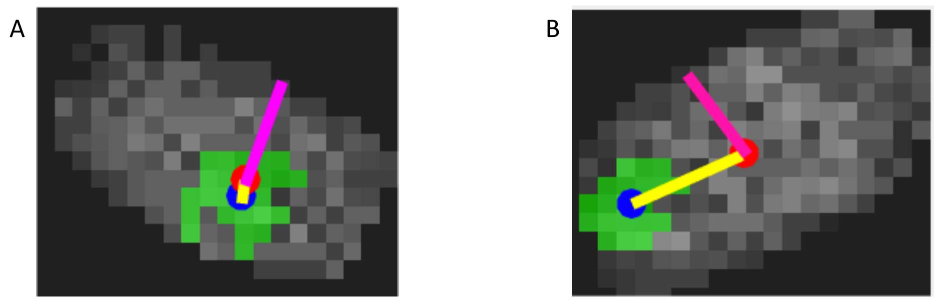

Figure 2.4: Representation of how the foci location is determined. Blue dot: foci centroid, red dot: cell centroid, pink line: cell radius, yellow line: distance between the two centroids. If the distance is higher than the radius, the foci is at the poles, otherwise it's at the spectal area. A. Foci at septal area. B. Foci at pole.

2.4. Transformation using *B. subtilis* 168

The pDR111 plasmid [Supplementary Figure A.9] served as the backbone for constructing two plasmids, namely pDR111_yqhY_GFP2_AmyE and pDR111_yqhY_GFP2_noAmyE. This modification was necessary because the origin present in pBbE5k is incompatible with *Bacillus subtilis*. The term "AmyE" refers to the alpha amylase gene, utilized for genome integration. In the pDR111_yqhY_GFP2_AmyE case, the yqhY_GFP2 sequence is integrated into the B. subtilis genome within the AmyE gene. This ensures genome integration and the coexistence of both tagged and untagged YqhY within the bacteria, maintaining wildtype functionality while tagging the protein of interest. This design also mitigates potential interference from the GFP molecule. Conversely, for pDR111_yqhY_GFP2_noAmyE, the yqhY_GFP2 sequence is integrated into the B. subtilis genome within the YqhY gene, resulting in the presence of only tagged YqhY. The exclusive presence of tagged YqhY offers advantages such as increased GFP intensity and enhanced control over protein expression, allowing manipulation and induction based on experimental requirements. This approach aligns with previous methods which used only tagged YqhY, providing a higher likelihood of reproducing similar results.

2.4.1 Cloning of pDR111_yqhY_GFP2_AmyE in DH5α

The yqhY_GFP2 sequence was inserted in the construct using Gibson Assembly. Polymerase chain reaction (PCR) was used to obtain most of the desired gene constructs. The specifications of the used plasmid can be found in Figure 2.5.

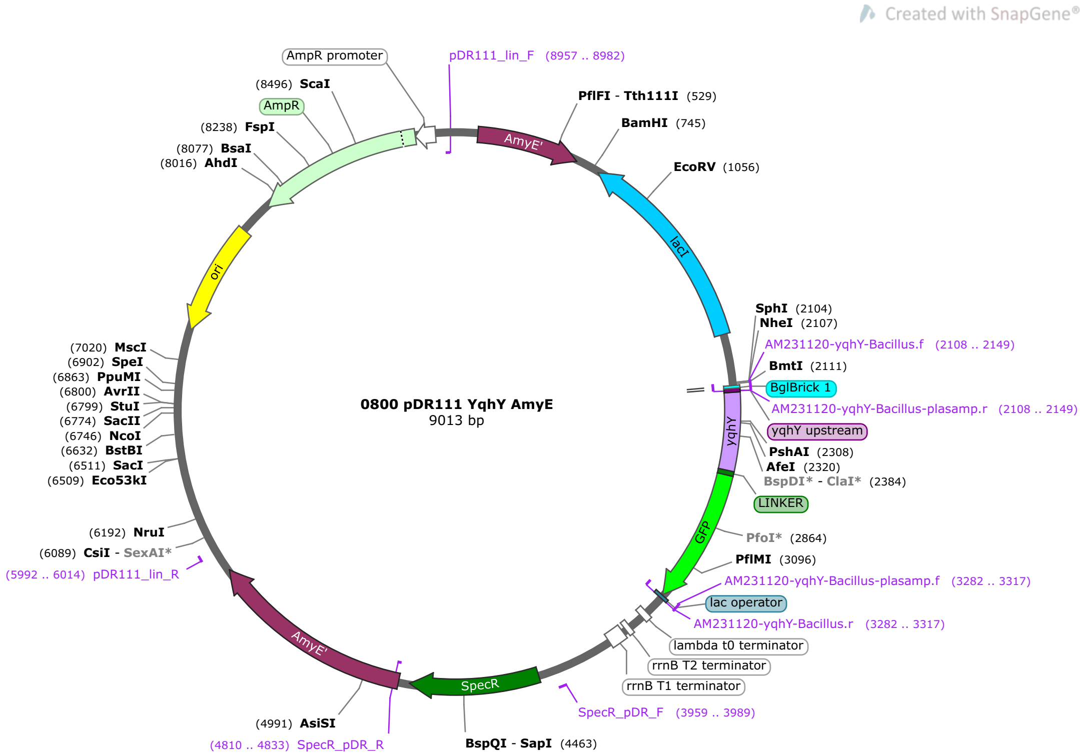

Figure 2.5: pDR111_yqhY_GFP2_AmyE map created with *SnapGene*. It presents yqhY_GFP2 plus an origin (ori), resistance to both Ampicillin (AmpR) for *E. coli* and Spectinomycin (SpecR) for *B. subtilis*, alpha amylase (AmyE) for genome integration and an IPTG inducible site (lacI).

Cloning was performed following the same protocols explained in Section 2.2. For PCR amplification of the insert forward primer AM231120-yqhY-Bacillus.f and reverse primer AM231120-yqhY-Bacillus.r were used. For PCR amplification of the vector forward primer AM231120-yqhY-Bacillus-plasamp.f and reverse primer AM231120-yqhY-Bacillus-plasamp.r were used [Table 2.5].

Table 2.5: Primers used for Gibson Assembly to form pDR111_yqhY_GFP2_AmyE.

Primer	Oligonucleotide sequence 5'- 3'
AM231120-yqhY-Bacillus-plasamp.f	CAGTGGCGGTTCTTAA taattgttatccgctcacaa
AM231120-yqhY-Bacillus-plasamp.r	ccgtaaaaaagatcttttgaattc AGCTTAGTCGACAGCTAG
AM231120-yqhY-Bacillus.f	CTAGCTGTCGACTAAGCT gaattcaaaagatcttttttacgg
AM231120-yqhY-Bacillus.r	ttgtgagcggataacaatta TTAAGAACCGCCACTG

PCR was performed according to a specific program [Table 2.6]. Specifically, the elongation time was set according to the length of the plasmid; 30 seconds for each kbp.

Table 2.6: PCR program amplification pDR111_yqhY_GFP2 with primers with melting temperature 58 °C.

Stage	Stage 1 (x1)	Stage 2 (x25)			Stage 3 (x1)	
Temperature	98 °C	98 °C	58 °C	72 °C	72 °C	4 °C
Time (min)	1.00	0.15	0.30	4.30	7.00	∞
Step	Activation	Denaturation	Annealing	Extension	Final Extend	Collection

Gibson assembly was performed following the protocol explained in section 2.2.4. In this case, the insert is 1210 bp, and the vector 7803 bp, which means that the amount of insert needed for 100 ng of vector is 46.5 ng. Transformation and inoculation were performed as previously explained, with the only difference being the use of ampicillin instead of kanamycin. Colony PCR was not performed, and the whole plasmid was sequenced to check for mutations.

2.4.2 Cloning of pDR111_yqhY_GFP2_noAmyE in DH5α

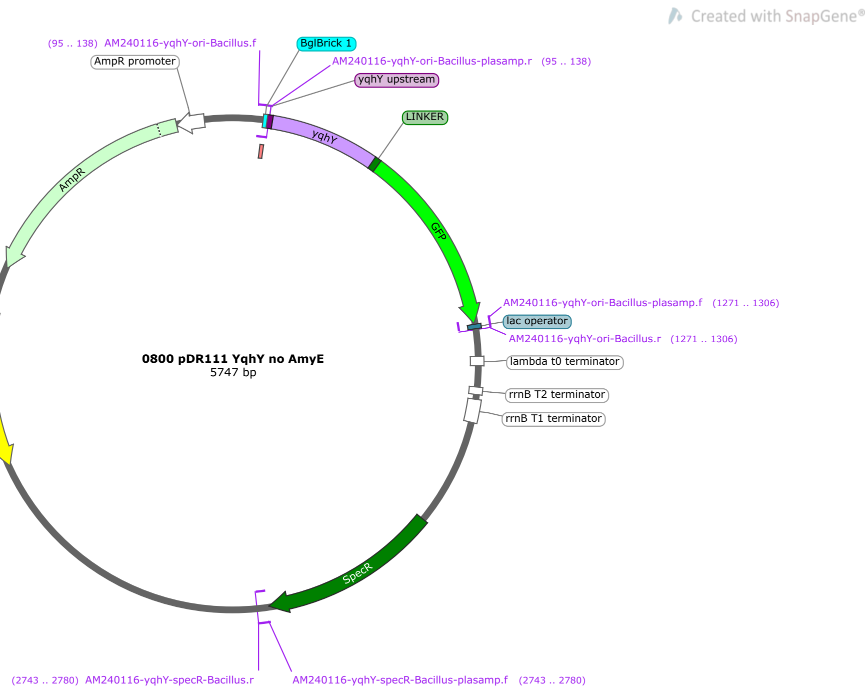

Figure 2.6: pDR111_yqhY_GFP2_noAmyE map created with *SnapGene*. It presents yqhY_GFP2 plus an origin (ori), resistance to both Ampicillin (AmpR) for *E. coli* and Spectinomycin (SpecR) for *B. subtilis*, and an IPTG inducible site (lacI).

The yqhY_GFP2 sequence was inserted in the construct using a three insert Gibson Assembly. pDR111 was used to create two segments, one containing the origin (ori) and the ampicillin resistance (AmpR), and the other with the spectinomycin (SpecR). A third insert was generated from pBbE5k_yqhY_GFP2 to obtain the yqhY_GFP2 sequence. PCR was used to obtain the desired gene constructs. The specifications of the used plasmid can be found in Figure 2.6. Cloning was performed following the same protocols explained in Section 2.2. Three segment were amplified using PCR. The insert including the ori and the AmpR was amplified using the forward primer 237955406 and reverse primer 237955404 were used. For yqhY_GFP2 the forward primer 237955401 and reverse primer 237955402 were used. Finally, for SpecR the forward primer 237955403 and reverse primer 237955405 were used [Table 2.7].

Table 2.7: Primers used for Gibson Assembly between pBbE5k_yqhY_GFP2 and pDR111 to form pDR111_yqhY_GFP2_noAmyE.

Primer	Oligonucleotide sequence 5'- 3'
237955401	TCAAATAAGGAGTGTCAAGA gaattcaaaagatctttttttacgg
237955402	ttgtgagcggataacaatta TTAAGAACCGCCACTG
237955403	cagtggcggttcttaa TAATTGTTATCCGCTCACAA
237955404	CCGTAAAAAAAGATCTTTTGAATTC tcttgacactccttatttga
237955405	ttgcagggtatgtttctc TGTTTCCACCATTTTTTCAA
237955406	ttgaaaaaaatggtggaaaca GAGAAACATACCCTGCAA

PCR was performed according to a specific program [Table 2.8]. Specifically, the elongation time was set according to the length of the plasmid; 30 seconds for each kbp.

Table 2.8: PCR program amplification pDR111_yqhY_GFP2_noAmyE with primers with melting temperature 58 °C.

Stage	Stage 1 (x1)	Stage 2 (x25)			Stage 3 (x1)	
Temperature	98 °C	98 °C	58 °C	72 °C	72 °C	4 °C
Time (min)	1.00	0.15	0.30	1.30	7.00	∞
Step	Activation	Denaturation	Annealing	Extension	Final Extend	Collection

Gibson assembly was performed following the protocol explained in section 2.2.4. In this case, the inserts are 1205 bp (yqhY_GFP2) and 1475 bp (SpecR), while the vector 3067 bp, which means that the amounts of insert needed for 100 ng of vector are 118 ng and 144 ng respectively. Transformation and inoculation were performed as previously explained, with the only difference being the use of ampicillin instead of kanamycin. Colony PCR was not performed, and the concentration was measured using Qubit Fluorometer (Invitrogen by ThermoFisher, #Q33238), following the $Qubit^{TM}$ dsDNA BR Assay Kit (#Q32850) protocol. Subsequently, the whole plasmid was sequenced to check for mutations using plasmidsaurus.

2.4.3 Mediums

MM competence medium

The MM competence medium is composed of:

- 10 mL SMM solution
- 0.1 mL of 50% glucose
- 0.1 mL tryptophan solution [2 mg/mL]
- 0.06 mL $MgSO_4$ [1 M]
- 0.01 mL of 20% CAS
- 0.005 mL of 0.22 % $Fe - NH_4 - citrate$

Starvation medium

The starvation medium is composed of:

- 10 mL SMM solution
- 0.1 mL sol E = 50 % glucose
- 0.06 mL $MgSO_4$ [1 M]

SMG medium
The SMG medium is composed of:

- 100 mL SMM medium
- 1 mL 50% (w/v) glucose
- 1 mL 10% (w/v) glutamate
- 1 mL 0.2% (w/v) L-tryptophan

SMM medium
The SMM medium is composed of:

- 200 mL 5x SM base medium
- 10 mL 100x Trace element solution
- 10 mL 100x Fe-Citrate-Solution
- 10 mL of 50% Glucose
- 10 mL Tryptophan [5 mg/mL] for *B. subtilis 168*
- N-source solution to final concentration of 1 %
- 1,000 mL deionized H_2O

5x SM base medium:
The 5x SM base medium is composed of:

- 175 g K_2HPO_4
- 75 g KH_2PO_4
- 12.5 g $Na_3 - Citrate$ Σ 2 H_2O
- 2.5 g $MgSO_4$ Σ 7 H_2O
- 2,000 mL deionized H_2O

N-source solutions:
The N-source solution is composed of:

- Glutamine 4% (do not autoclave)
- Glutamate 40%
- $(NH_4)_2SO_4$, with final concentration 0.2%.

100x Trace element solution:
The 100x Trace element solution is composed of:

- 0.55 g $CaCl_2$
- 0.1 g $MnCl_2$ Σ 4 H_2O
- 0.17 g $ZnCl_2$
- 0.033 g $CuCl_2$ Σ 2 H_2O
- 0.06 g $CoCl_2$ Σ 6 H_2O
- 0.06 g Na_2MoO_4 Σ 2 H_2O
- 1,000 mL of deionized H_2O

Do not autoclave, use sterile filtration!

100x Fe-Citrate-Solution:
The 100x Fe-Citrate-Solution is composed of:

- 0.0135 g $FeCl_3$ Σ 6 H_2O
- 0.1 g $Na_3 - Citrate$ Σ 3 H_2O
- 100 mL of deionized H_2O

Do not autoclave, use sterile filtration!

2.4.4 Transformation protocol in *B. subtilis*

To transform pDR111_yqhY_GFP2_AmyE and pDR111_yqhY_GFP2_noAmyE in *B. subtilis* 168, from a fresh plate (or glycerol stock) colonies were inoculated in 4 mL of MM competence medium, cultivating overnight at 37°C. To ensure high efficiency and consistency, multiple independent precultures were grown and combined the following morning. 10 mL of pre-warmed MM competence medium were inoculated with 600 μL of the preculture and incubated for 3 hours at 37°C until reaching an OD of approximately 0.3-0.4. 10 mL of prewarmed starvation medium were added and incubated for 1 hour at 37°C with agitation. Subsequently, 10 μL of DNA ($\approx$ 50 ng/μl) were roughly mixed with 400 μL of competent cells in an Eppendorf tube, shaking for 2 hours at 37°C. LB plates with spectinomycin (SpecR) (200 μL per petri dish) were used and incubated at 37°C.

2.4.5 Colony PCR

To check whether the colony had the insert or not, colony PCR was performed. For colony PCR, a ThermoFisher Scientific DreamTaq Green PCR Master Mix (2X) mastermix was used. This is a ready-to-use solution containing DreamTaq DNA Polymerase, optimized DreamTaq Green buffer, $MgCl_2$, and dNTPs. A colony was taken from the plate, put it in 50 μL of MilliQ water (PCR tube), and resuspend a bit (doesn't always nicely work, sometimes colony cohesion is super high). PCR machine was set to 95 °C for 3-5min, the tube was then mixed (2s vortex, or just pipette up and down a few times), and finally 1 μL of that lysate was added to the PCR reaction.

Two sequencing primers were used for pDR111_yqhY_GFP2_AmyE, starting from the beginning of the AmyE gene and ending at the end of the AmyE gene [Supplementary Figure A.10], respectively AmyE-full-up-F and AmyE-full-down-R [Table 2.9].

Table 2.9: Primers used for Colony PCR for pDR111_yqhY_GFP2_AmyE

Primer	Oligonucleotide sequence 5'- 3'
AmyE-full-up-F	TTCTGACGTCTTAATTAAGAAGACTACAGGAGGGCTGCGGCA TCCGGAATGCTCATGCCG
AmyE-full-down-R	CTCGAGGGATCCGAATTCGAAGACTTGGTACCTAAGAAAATG CCGTCAAATCCGCTCGCCA

Per reaction 10 μL of the 2X mastermix, 0.5 μL of forward primer (10 μM), 0.5 μL of reverse primer (10 μM), 1 μL of resuspended colony, and 8 μL Milli-Q were assembled in a total volume of 20 μL. A control group was used to check the results. The PCR was run using a thermocycler according to the program displayed in Table 2.10. Specifically, the elongation time was set according to the length of the plasmid; 1 minute for each kbp.

Table 2.10: Colony PCR program amplification of the Gibson Assembly's colonies with melting temperature 68 °C.

Stage	Stage 1 (x1)	Stage 2 (x35)			Stage 3 (x1)	
Temperature	95 °C	95 °C	60 °C	72 °C	72 °C	4 °C
Time (min)	18.00	0.30	0.30	6.00	5.00	∞
Step	Activation	Denaturation	Annealing	Extension	Final Extend	Collection

After the PCR run, the reactions were visualized using gel electrophoresis. The same gel composition as the Phusion PCR was used, see Section 2.2.2. All micro-wells were filed with 15 μL of sample and 4 μL of DNA BenchTop Ladder. The green DreamTaq mastermix already contains the loading dye.

2.5. Microscopy using *B. subtilis* 168

2.5.1 Microscope Settings

To analyze YqhY in *B. subtilis* cells, widefield fluorescence microscopy was used. An Olympus IX81F-ZDC2 microscope was used for the images with a 100x oil objective. The microscope had the following overall settings; a temperature of 37 °C, an EM gain of 200. For the fluorescent images a wavelength of 450 nm was used, the intensity was 50% and the exposure 50 milliseconds (ms).

2.5.2 Localization of YqhY

The strain must be grown in liquid culture in order to image the expression of YqhY. 4 μL of SpecR and 4 mL of LB media were added to a sterile colony tube and mixed. Using a pipette tip, a colony was streaked from the desired plate and put into the tube. Until the liquid culture tube attained an OD of 0.3–0.4, it was incubated at 37 °C and 180 rpm.

2.5.3 Preparation of cover slips

For the investigation under the fluorescent microscope, agarose cover-slips were utilised. The content of the pads was 1 μL SpecR, 1 μL glucose solution (50%), 1 μL glutamate solution (10%), 1 μL L-tryptophan solution (0.2%), 500 μL of SMM-based medium, and 500 μL 2x agar (1.5%). To make a sandwich, 300 mL of agar were pipetted on top of one cover-slip (epredia, 15 x 15 mm), and then another cover-slip was put on top of the agar. The cover-slip were left to dry for roughly twenty minutes. One of the cover-slips was then taken off, and 1 μL of the (induced) cultures was pipetted on top of the agar. To keep the cover-slip from drying out, it was positioned with the agar side facing the microscope glass slide (VMR, 22 x 50 mm) and the edges sealed with clear nail polish [Figure 2.5.3].

3 Results

The goal of the research is to understand how YqhY regulates ACC activity and plays a specific role in keeping the balance between other cellular needs and fatty acid production. Employing Green Fluorescent Protein (GFP) tagging, YqhY was tracked within bacterial systems, allowing fluorescence-based observation across diverse conditions. *E. coli* was utilized at first for plasmid production due to its suitability for genetic manipulation and transformation, but the experiments were subsequently attempted in *B. subtilis* for comparative analysis.

3.1. YqhY forms foci at the septum in *E. coli*

In the initial phase of our investigation, our primary objective was to replicate previously conducted experiments in order to validate earlier observations, specifically concerning the localization of YqhY expression predominantly at the poles. To achieve this, a critical step involved the successful fusion of yqhY_GFP2 in the pBbE5k_PlsB_msGFP2 construct, accomplished through the efficient and precise technique of Gibson Assembly in *E. coli* DH5α. The plasmid was then transformed in *E. coli* NCM 3722 since the strain is ideal for expression experiments.

The investigation into NCM 3722 pBbE5k_yqhY_GFP2 (not induced) provided intriguing insights. The expression of yqhY_GFP2 was visible, yet the distinct foci within many cells appeared relatively reduced and faint when compared to the broader cellular background. This expression level, when compared to existing literature [Figure 3.1.A], showcased a considerable decrease [Figure 3.1.B and 3.1.C]. This discrepancy may be attributed to the use of reference literature that focused on *B. subtilis*, a species characterized by high expression levels of YqhY [7]. In contrast, our experiment employed *E. coli*, which lacks the native gene counterpart in its genome and relies on the expression introduced through the plasmid. Furthermore, in this initial experiment, the gene was not induced, implying that the observable fluorescence solely results from leaky expression.

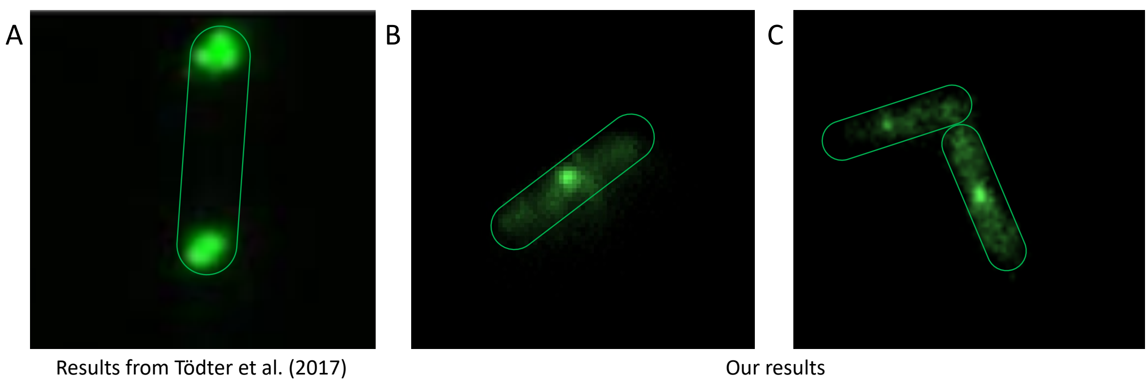

Figure 3.1: Expression of yqhY_GFP2 in NCM3722. A. Results from Tödter et al. in *B. subtilis* [7], B. and C. Two examples of bacteria showcasing dim foci in the septal area of the cells.

What set this observation apart was the unexpected localization pattern of these foci. Unlike prior documented instances, the foci exhibited a robust localization in the septal area of the cells, diverging from the anticipated polar localization reported [Figure 3.1.A and Figure 3.2.B] [7].

To validate our findings, we performed a rigorous analysis using MicrobeJ, an ImageJ plugin, and MatLab. The quantitative analysis revealed a consistent trend, with the program identifying 1, 2, or 3 foci in 85% of the cultured samples [Figure 3.2.A]. This alignment between qualitative and quantitative outcomes confirms previous research stating that YqhY forms foci in bacteria.

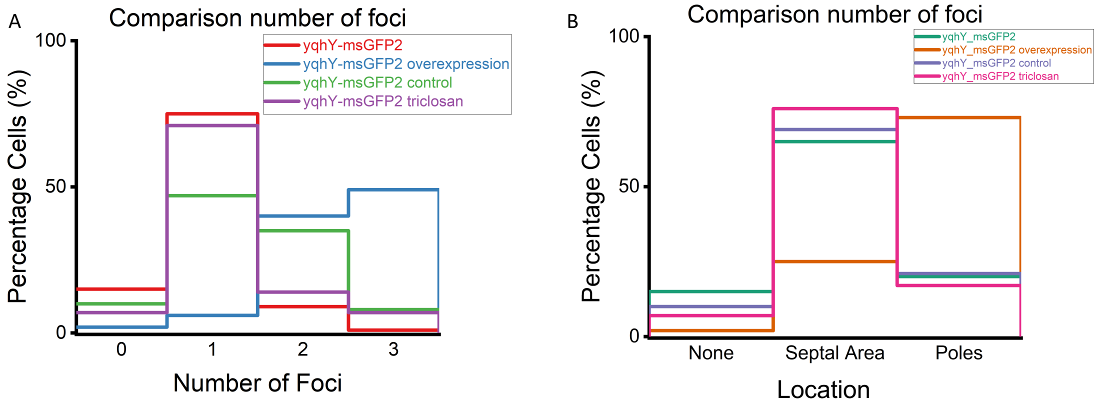

Figure 3.2: Quantitative foci analysis results. A. Results showcasing percentage of cells with certain number of foci. Four conditions compared: NCM3722 pBbE5k_yqhY_GFP2 (red), NCM3722 yqhY_GFP2 overexpression (blue), NCM3722 pBbE5k_yqhY_GFP2 control (green), and NCM3722 pBbE5k_yqhY_GFP2 triclosan (purple). B. Results showcasing percentage of cells at a certain location with each cell. Four conditions compared: NCM3722 pBbE5k_yqhY_GFP2 (green), NCM3722 yqhY_GFP2 overexpression (orange), NCM3722 pBbE5k_yqhY_GFP2 control (purple), and NCM3722 pBbE5k_yqhY_GFP2 triclosan (pink).

The divergence in foci localization observed between *E. coli* and *B. subtilis* is intriguing. Initial hypotheses pointed towards differences in membrane lipid composition as a potential explanation for the observed shift. The consideration of phospholipid headgroup composition as a potential explanation for the observed divergence in foci localization between *E. coli* and *B. subtilis* is driven by the distinctive properties of phospholipids in these bacteria. Notably, phosphatidylethanolamine (PE) is zwitterionic, containing both positive and negative charges. On the other hand, cardiolipin (CL) and phosphatidylglycerol (PG) are negatively charged. The poles and septum of *E. coli* are thought to accumulate PG/CL. These electrostatic differences in headgroup composition could influence the localization of cellular foci. Unfortunately, despite presenting differences; *E. coli* exhibiting a lipid profile with 5% CL, 75% PE, and 20% PG [36], and *B. subtilis* displaying a notably lower percentage of PE (around 50%) and a higher concentration of PG (approximately 45%) [37], the lipid composition within the foci and septal areas of both bacteria appeared surprisingly similar [38]. This observation suggests that factors beyond lipid composition may contribute to the observed shift, for example it might be influence by interactions with other proteins.

3.2. Overexpression of YqhY using IPTG in NCM3722

Given the absence of the protein in its genome and the lack of protein induction, *E. coli* struggles in replicating the conditions observed in *B. subtilis*. With *B. subtilis* showcasing high expression levels of the protein, approximately 22,500 molecules per cell, efforts were made to mimic this concentration through the overexpression of YqhY in *E. coli*.

3.2.1 Overexpressing YqhY results in a localization shift

With the application of IPTG, we aim to trigger the expression of YqhY, anticipating the observation of foci with increased intensity. This expectation is based on the precedent results by Todter et al.'s article, where foci displayed a greater intensity [Figure 3.3.A][7].

Unexpectedly, the first time we conducted the experiment the outcome of the YqhY overexpression revealed a notable shift in behavior. While there was a clear and significant increase in fluorescence intensity under microscopic observation, an unexpected outcome emerged: the complete loss of its usual localization pattern [Supplementary Figure A.11]. This outcome contrasts sharply with both the anticipated localization described in existing literature and the subtle localization witnessed earlier.

The drastic change to a fully uniform distribution could be attributed to various factors. Firstly, it is crucial to emphasize the divergence in bacterial characteristics. The outcomes observed in *B. subtilis* may not necessarily be replicated in *E. coli* due to inherent physiological, structural, and behavioral distinctions between these two organisms. If the heterogeneity in bacteria is not the cause for this disparity, an alternative explanation could be that there is an overabundance of YqhY in *E. coli*, leading to an excess of this molecule. Within our hypothesis, we posit that YqhY engages in either direct or indirect interactions with ACC. Should this scenario hold true, the concentrations of both proteins might influence their respective localizations. In *B. subtilis*, there are approximately 22,500 molecules per cell of YqhY and around 14,000 molecules per cell of each ACC subunit [18], indicating an approximate ratio of one and a half molecules of YqhY per ACC. However, in *E. coli*, there are only around 10,000 molecules per cell of each ACC subunit [39]. When YqhY is overexpressed in *E. coli*, it might lead to an abundance of molecules per cell, exceeding the physiological ratio of 1.5 molecules of YqhY per ACC. The surplus YqhY molecules might fail to localize at the foci since they cannot all interact with ACC. Consequently, the excess YqhY molecules disperse into the cytoplasm, resulting in the overall fluorescence of the entire bacteria.

In order to validate the reliability of our findings and account for any potential confounding factors, we conducted the same experiment once again. This approach was used to ascertain whether the observed shift in behavior was attributable to the experimental conditions or if it indeed reflected the behavior of YqhY.

Upon conducting the repeated experiment, we were gratified to observe that the outcomes aligned precisely with our initial expectations. The bacteria displayed a notable concentration of high-intensity foci at the poles [Figure 3.3.B and 3.3.C], mirroring the distinctive pattern documented in the Todter et al. paper that served as a foundational reference for our research [Figure 3.3.A][7]. This replication of the expected results not only validates the robustness of our experimental design but also reinforces the consistency of the observed protein behavior.

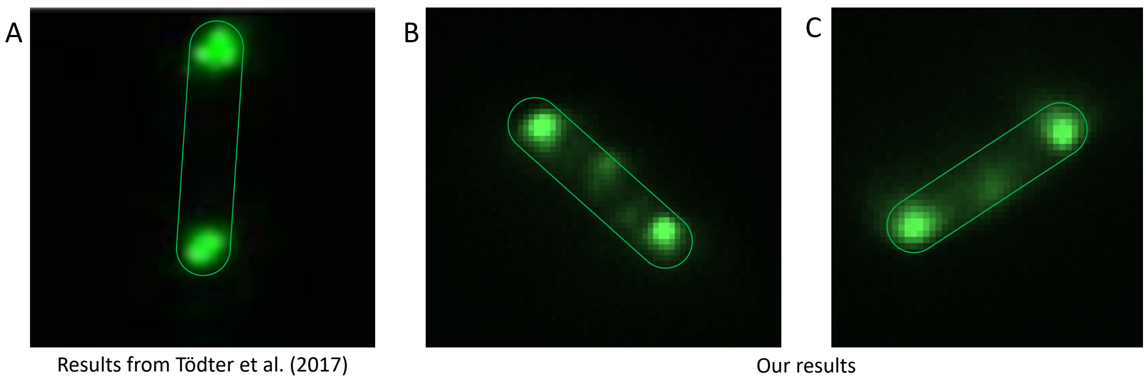

Figure 3.3: Foci formation of yqhY_GFP2 after overexpression induced in IPTG in NCM3722. A. Results from Todter et al paper [7]. B. Example of high intensity foci formation at the poles. C. Example of high intensity foci formation at the poles.

To validate the qualitative results and ensure the absence of overlooked information, a quantitative analysis was conducted. The quantitative assessment revealed 1, 2, 3 or 4 foci in 97% of the cases [Figure 3.2.A]. Furthermore, around 70% of the foci were located at the poles. This is a notable shift compared to the uninduced results, which showcased foci mainly at the septal area. Finally, there was a visible discrepancy in foci intensity between the foci at the septal area and poles [Supplementary Figure A.12].

3.2.2 Growth rate unvaried when YqhY is overexpressed

The goal of this experiment was to compare the growth rates of two different sets of bacterial cultures: one that had YqhY overexpressed using IPTG and the other that had uninduced YqhY. This study was motivated by the possibility of YqhY influencing the fatty acid pathway. Since YqhY is thought to inhibit ACC in *B. subtilis* [7], an overexpression of this protein should result in less fatty acid synthesis and, in turn, slower growth. It's crucial to note that the protein under scrutiny, YqhY, originates from *B. subtilis* but is expressed in *E. coli*, introducing the possibility of anomalous behavior due to the different bacterial hosts. To test this hypothesis, overexpression of YqhY was carried out, and the growth rates of two sets of cultures were monitored – one with unexpressed YqhY and the other with YqhY overexpressed.

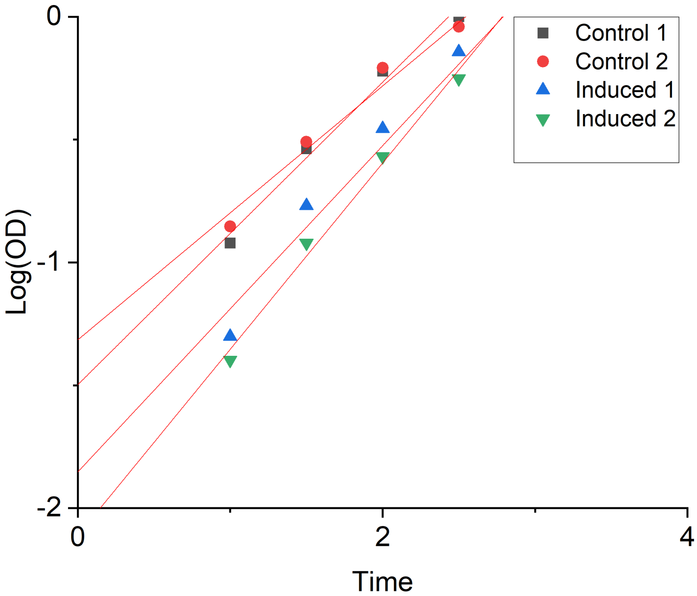

Figure 3.4: Growth rate monitored by measuring the optical density (OD) every doubling time (30 minutes). Two conditions were tested; unexpressed YqhY as a control (black), and overexpressed YqhY using IPTG (red). Both induced cultures resulted in a decrease in rate, with Induced 2 showcasing a significant reduction.

In examining the growth rates under the two conditions, it is noteworthy that the overall growth patterns remain relatively consistent [Figure 3.4]. The slopes across all four solutions exhibit a uniform trend, suggesting that the alterations induced by the experimental conditions do not significantly impact the overall growth rate. Intriguingly, the primary distinction lies in the starting points of the growth curves. Specifically, for the two induced colonies [Figure 3.4, Induce 1 and 2], the initial growth rate is comparatively lower than in the non-induced conditions. This discrepancy in starting points implies that the induction process may influence the initial stages of bacterial growth rather than the overall growth rate, highlighting a nuanced aspect of the experimental outcomes. The decision to refrain from repeating the experiment in *E. coli* was based on the organism of interest being *B. subtilis*. Opting to conduct the experiment directly with *Bacillus* was deemed more relevant.

3.3. Exposure to antibiotics

Given that YqhY is thought to be involved in the fatty acid pathway, we conducted an experimental study aimed at understanding the impact of fatty acid impairment on the localization of YqhY within bacterial cells. To inhibit the fatty acid pathway, we exposed the bacteria to two distinct antibiotics: cerulenin and triclosan. Cerulenin disrupts lipid synthesis in bacteria, specifically inhibiting the enzyme beta-ketoacyl-acyl carrier protein (ACP) synthase, also known as FabF. On the other hand, triclosan operates by inhibiting bacterial enoyl-acyl carrier protein reductase (FabI). By subjecting the bacteria to these antibiotics, we sought to elucidate how alterations in lipid metabolism, resulting from the specific modes of action of cerulenin and triclosan, might influence the subcellular localization of YqhY. Notably, this approach aligns with previous work in Greg Bokinsky's lab, where a similar study was performed on PlsB, the sole regulator of membrane synthesis in *E. coli* [15].

Expanding upon earlier studies, Greg Bokinsky's group investigated the localization of PlsB in living cells using Green Fluorescent Protein 2 (GFP2) labelling. This labelling technique revealed discrete fluorescent foci at the poles. These foci, thought to be filaments similar to those seen in previous research by Wilkison et al. [40], were shown to be the inactive form of PlsB. *E. coli* was exposed to triclosan, which led to the disappearance of all foci. This is due to the fact that inhibition of fatty acid synthesis, specifically FabI, leads to a depletion of phospholipids available for membrane synthesis in bacteria. In response to this deficit, PlsB becomes activated in an attempt to enhance phospholipid production and counteract the impact of the inhibition. As shown in Figure 3.5, as PlsB undergoes activation, its localization transitions from foci at the poles to the cytoplasmic region, adding a compelling layer to the hypothesis that PlsB becomes monomeric when in an active state.

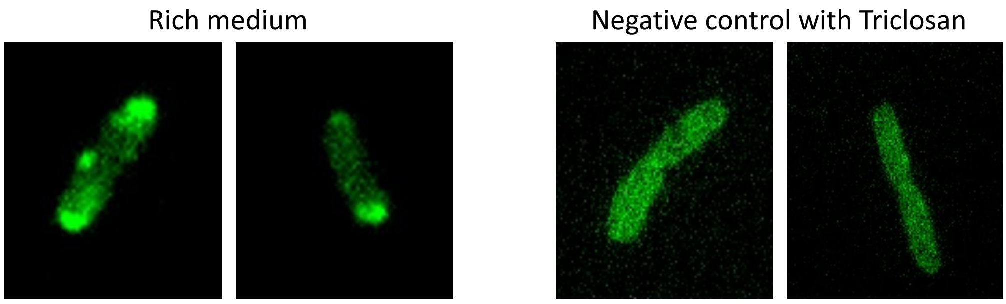

Figure 3.5: Upon labeling of the enzyme PlsB with GFP2, foci concentrated at the cell poles of *E. coli* are formed (left). When fatty acid synthesis is inhibited by triclosan these foci disappear, implying that the foci indeed are inactive PlsB filaments. Adapted from unpublished work by J. Beije, Bokinsky Lab (2022).

3.3.1 Growth with fatty acid synthesis inhibitors is inconclusive

The aim of this experiment was to investigate how the subcellular localization of YqhY is affected by inhibitors of fatty acid synthesis, specifically triclosan and cerulenin. Previous research has proposed that YqhY acts as an inhibitor of ACC, influencing fatty acid synthesis [7]. According to our theory, YqhY inhibits ACC and localizes at the foci during this process. Upon the introduction of triclosan, FabI is inhibited, disrupting the normal progression of fatty acid synthesis. In *E. coli*, this inhibition results in impaired conversion of enoyl-ACP to acyl-ACP, a crucial molecule for fatty acid elongation. The diminished production of acyl-ACP due to FabI inhibition leads to a decrease in holo-ACP formation. Holo-ACP serves as a carrier molecule for malonyl groups during fatty acid biosynthesis. It's noteworthy that ACP is a dynamic protein, and the acyl group carried by Acyl-ACP can be recycled to produce apo-ACP, eventually reforming holo-ACP through further modification. However, when FabI is inhibited, the concentration of Acyl-ACP is reduced, consequently lowering holo-ACP levels. This reduction in holo-ACP availability leads to the accumulation of malonyl-CoA, since it cannot efficiently be converted to malonyl-ACP. The increased presence of malonyl-CoA results in reduced activity of ACC, requiring YqhY to enhance its inhibitory effect. This leads to a higher concentration of YqhY at the foci, resulting in increased intensity. Additionally, since triclosan and cerulenin are known to impact bacterial growth, exposure to these antibiotics is expected to reduce the bacterial population. In order to do this, three cultures were prepared: a negative control, one treated with triclosan, and another with cerulenin. After a 30-minute incubation with the antibiotics, the cultures were placed on agar pads containing the respective antibiotics.

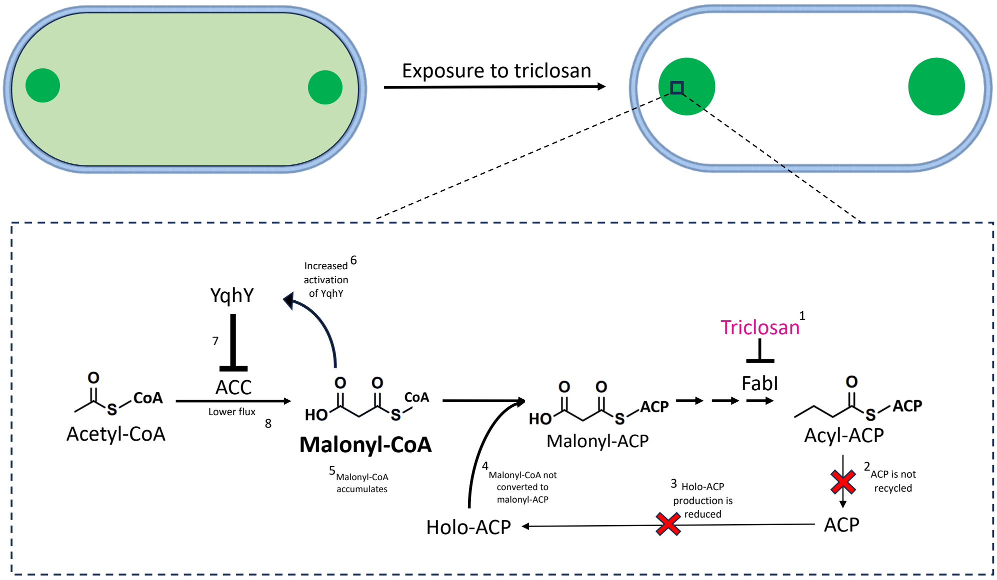

Figure 3.6: Diagram showcasing our theory regarding triclosan exposure in steps. 1. Once triclosan enters the system, FabI gets inhibited. 2. The inhibition of FabI causes the fatty acid biosynthesis pathway to halt, leading to a decrease in Acyl-ACP production, and therefore, a lower recylcing rate of ACP. 3. Since ACP is not recycled, holo-ACP production is reduced 4. Without holo-ACP, malonyl-CoA cannot be converted into malonyl-ACP 5. Malonyl-CoA accumulates. 6. The accumulation of malonyl-CoA sends a signal to YqhY to increase its inhibitory activity. 7. ACC is further inhbited by YqhY. 8. Acetyl-CoA is not converted into malonyl-CoA.

Upon imaging, the negative control exhibited no foci [Figure 3.7.A and 3.7.D], even after thorough examination using different areas on the pad and attempting Z-stacks to ensure comprehensive observation, contradicting literature [7] and results described in section 3.1 [Figure 3.7.C]. Conversely, the cells exposed to triclosan displayed a clear presence of foci. These foci appeared faint and were primarily located in the septal region rather than the poles of the cells [Figure 3.7.E]. It is worth noting that the pad containing cells exposed to triclosan displayed fewer bacteria compared to the control as expected [Figure 3.7.B]. Unfortunately, attempts to image cells exposed to cerulenin were unsuccessful.

Notably, when comparing the control [Figure 3.7.D] with the results obtained for the localization experiment [Figure 3.7.C], the two behaviors are very different. This inconsistency poses a perplexing challenge, as the bacteria showcased no foci at all. While the possibility of cell or slide mix-up was considered, both researchers involved are confident this did not occur. However, the age of the plate, being four weeks old, could explain the atypical behavior observed in the negative control. This aging may have affected the bacterial response to the antibiotic or may have influenced YqhY expression and localization, introducing an additional variable.

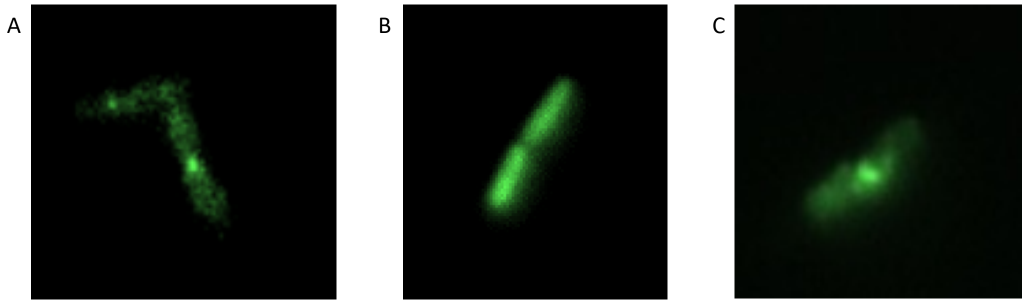

Figure 3.7: Comparative results between yqhY_GFP2 control and exposure to triclosan in NCM3722. A. Magnification 30x of fluorescent image of yqhY_GFP2 showcased in section 3.1. B. Magnification 30x of fluorescence image of control. C. Magnification 30x of fluorescence image of bacteria exposed to triclosan.

Likewise for these experiments, a quantitative methodology was adopted to precisely assess the outcomes. Specifically, the control group, consisting of NCM 3722 with YqhY tagged with GFP, exhibited a noteworthy 90% occurrence of foci [Figure 3.2.A, green]—a finding that distinctly contrasted with the qualitative analysis. Also in this case, the MatLab code may be taking heterogeneity in intensities into account instead of relevant foci, leading to false positives. Conversely, the triclosan results match with the qualitative observations, revealing that 87% of bacteria within this experimental condition presented foci [Figure 3.2.A].

The septal localization of foci in triclosan-exposed cells (80%, Figure 3.2.B), present in previously mentioned experiments [Section 3.1], remains intriguing and warrants further investigation to elucidate the underlying mechanisms governing this response.

3.3.2 Exposure to fatty acid synthesis inhibitors increases the foci intensity

Since we expected a higher intensity of the foci when in contact with triclosan, and since the control didn't exhibit any foci, we repeated the experiment. This time the cultures were grown to an O.D. of 0.3, and then exposed to pads containing the two antibiotics. Also in this case, a control experiment with normal pads was performed. The pads were continuously monitored over time using a widefield microscope, capturing snapshots at 2.5-minute intervals throughout a 30-minute duration.

Unexpectedly, both the control and triclosan-exposed samples revealed the emergence of foci after the fourth time point, indicating a significant shift approximately 10 minutes into the experiment [Figure 3.8]. The control exhibited dim foci [Figure 3.8.B], suggesting that in the previous iteration [Section 3.3.1], the lack of observable foci might have been a result of the bacteria adapting to the pad environment, necessitating additional time for the manifestation of focal points.

As anticipated, the triclosan-exposed samples showcased distinct foci with higher intensity compared to the control [Figure 3.8.D]. As shown in Figure 3.8.E, the average foci intensity of the control is around 4500 AU, meanwhile the average for the triclosan is around 7500 AU. These results contribute to the hypothesis that in presence of FabI inhibitor, YqhY has to increase its inhibitory activity on ACC, resulting in a higher intensity of the foci. Notably, a significant proportion of these foci localized in the septal area, aligning with previous experiments.

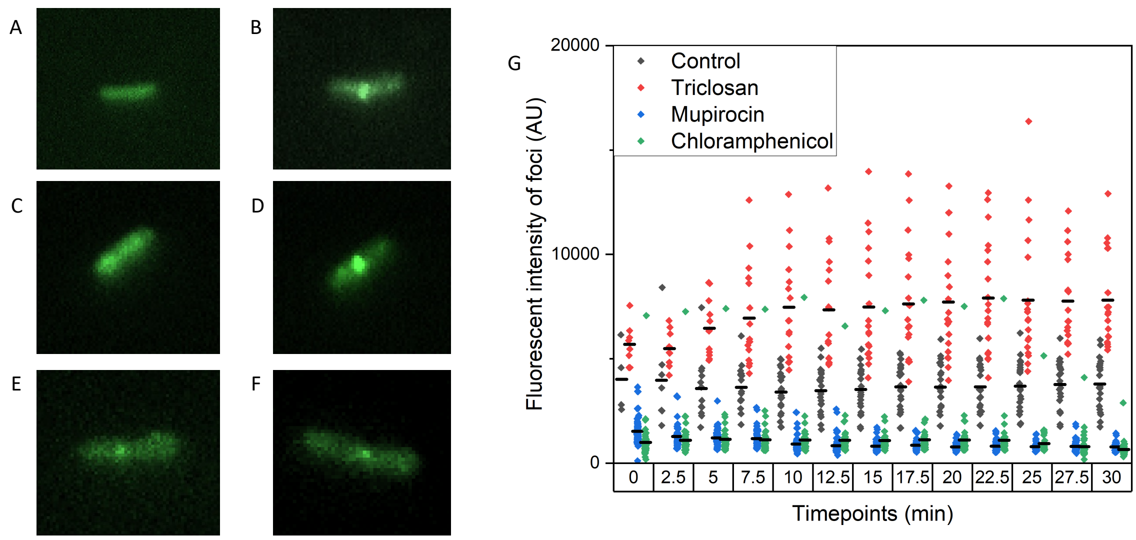

Figure 3.8: Comparative results between yqhY_GFP2 control and exposure to triclosan, chloramphenicol and mupirocin in NCM3722. A. Control at T1. No visible foci. B. Control at T13. Dim foci at the septal area C. Triclosan at T1. No visible foci, but higher intensity compared to the control. D. Triclosan at T13. Clear high intensity foci in the septal area. E. Chloramphenicol at T13. Dim foci F. Mupirocin at T13. Dim foci. G. Grouped column scatter with index data plot showing the average intensity of every foci as data points per time interval. Foci control: black dots, Foci triclosan: red dots, Foci mupirocin: blue dots, Foci chloramphenicol: green dots. Average intensity of all foci per time interval: black line.

3.3.3 Translation inhibitors do not have an effect on the foci intensity

After obtaining the results from the exposure to triclosan, we wanted to try and understand the underlying mechanisms contributing to the observed increase in the intensity of foci. We aimed to clarify whether this elevation was due to an increase in the concentration of malonyl-CoA resulting from the inhibition of FabI or if it was a consequence of a general halt in cell growth. To address this, we implemented chloramphenicol and mupirocin, potent inhibitors of protein translation.

Chloramphenicol is an antibiotic that exerts its inhibitory effect on protein translation by targeting the bacterial 50S ribosomal subunit. By binding to this region, chloramphenicol disrupts the catalytic activity of the ribosome, preventing the formation of peptide bonds and thereby halting the elongation of the growing polypeptide chain. On the other hand, Mupirocin, primarily targets the enzyme isoleucyl-tRNA synthetase, which is responsible for attaching the amino acid isoleucine to its corresponding transfer RNA (tRNA), a crucial step in the initiation of protein translation. Upon binding to the enzyme, mupirocin forms a stable complex, preventing the proper attachment of isoleucine to tRNA.

Overall, a halt in protein translation induced by chloramphenicol and mupirocin would result in a general shutdown of cell growth. If our hypothesis holds true, indicating that malonyl-CoA concentration is a pivotal factor influencing the inhibitory activity of YqhY, the outcomes of this experiment are anticipated to mirror the control conditions. In essence, this would suggest that the increase in intensity of foci is independent of the growth rate, reinforcing the connection between malonyl-CoA concentration and the observed inhibitory effects. To do this, a culture was grown to an O.D. of 0.3, and then exposed to two different pads containing the two antibiotics respectively. The pads were continuously monitored over time using a widefield microscope, capturing snapshots at 2.5-minute intervals throughout a 30-minute duration.

The observed results of bacteria exposed to both chloramphenicol and mupirocin, displaying low-intensity foci primarily at the septum [Figure 3.8.E and 3.8.F], align with the anticipated outcomes and provide support for the conclusions drawn from the triclosan experiment. This finding suggests that the previously observed increase in intensity may indeed be linked to a low concentration of malonyl-CoA rather than a cessation of cell growth.

As shown in Figure 3.8.G, the intensity for both antibiotics is lower compared to the control, around 1200 AU, which could be due to a lower concentration of YqhY in the sample. The decreasing number of foci with time may be indicative of a decrease in ACC to inhibit [Supplementary Figure A.13]. To gain further insights and confirm these behaviors, additional investigations using *B. subtilis* are recommended. This cross-species validation is crucial to ensuring that the patterns observed in *E. coli* are consistent and applicable to *B. subtilis*, thereby enhancing the robustness and reliability of the conclusions drawn from the experiment.

3.4. Localization of YqhY in *B. subtilis*

In the second phase of our investigation, our primary objective was to replicate previously conducted experiments in order to validate well-established observations, specifically concerning the localization of YqhY expression predominantly at the poles and triclosan's effects on *B. subtilis*. To achieve this, a critical step involved the successful insertion of the yqhY_GFP2 in pDR111, to create both pDR111_yqhY_GFP2_AmyE and pDR111_yqhY_GFP2_noAmyE using Gibson Assembly. The plasmid was then transformed in *B. subtilis* 168 since the strain is ideal for expression experiments.

Regrettably, despite diligent efforts, the construction of the required plasmids proved challenging within the constraints of our project timeline. The intricate nature of the molecular biology procedures and the complexity of the desired constructs posed unexpected hurdles. Unfortunately, these challenges, compounded by time constraints, compromised our ability to successfully visualize tagged YqhY as initially planned. Despite the setbacks, this experience provides valuable insights into the complexities of experimental design and execution, emphasizing the need for meticulous planning and resource management in future endeavors.

4

Conclusion and discussion

In conclusion, our study set out to unravel the precise role of YqhY in modulating ACC activity to maintain a delicate equilibrium between fatty acid synthesis and other cellular demands. Our initial hypothesis posited YqhY as an inhibitor of ACC, responsive to membrane signals, thereby influencing malonyl-CoA production. Despite challenges in pinpointing the specific regulatory mechanism, our investigations led to intriguing insights. We proposed a direct interaction between YqhY and ACC at the poles, supported by their tendencies to form foci in this cellular region [7][17]. We speculate that the focal form of YqhY corresponds to its inhibitory state, while an inactive state prompts its relocation to the cytoplasm, facilitating an increase in ACC activity—an observation paralleling the behavior of Asp23 in response to alkaline shock [20].

To dissect the intricate dynamics governing YqhY and its impact on ACC activity, we employed GFP tagging for fluorescence-based monitoring within bacteria. The experimental journey commenced in the well-characterized *E. coli*, chosen for its ease of manipulation, followed by attempted analysis in *B. subtilis*—the focus organism. Initial replication of experiments validated prior findings, confirming YqhY expression as foci. Surprisingly, contrary to previous research [7], the observed foci were predominantly located in the septal area rather than at the poles, challenging our anticipated subcellular localization pattern. These results were obtained in *E. coli* where YqhY normally is absent. Subsequent phases exposed bacteria to various stress-inducing scenarios, including YqhY overexpression, and antibiotic exposure. These conditions aimed to shed light on the dynamic responses of YqhY, unraveling its behavior under stress, and illuminating its role in modulating ACC activity. The collective findings contribute to advancing our understanding of the intricate interplay between YqhY and ACC, paving the way for future investigations into bacterial stress responses and regulatory mechanisms.

4.1. YqhY forms foci

The unexpected localization pattern of YqhY foci observed in our investigation, particularly the robust presence in the septal area of *E. coli* cells, diverges significantly from the anticipated polar localization reported in literature for *B. subtilis*. The initial aim of our investigation was to replicate established experiments that validated YqhY's predominant expression at the poles. The successful replacement of PlsB_msGFP2 with yqhY_GFP2 in the pBbE5k_PlsB_msGFP2 construct facilitated a precise examination of YqhY expression patterns in *E. coli* NCM 3722.

Interestingly, the analysis of NCM 3722 pBbE5k_yqhY_msGFP2 (not induced) revealed a visible yet reduced and faint presence of YqhY foci compared to literature references. This discrepancy could be attributed to the absence of the native gene counterpart in *E. coli*, leading to lower expression levels introduced through the plasmid. Furthermore, the unexpected localization of foci in the septal area does not align with prior documented instances of polar localization in *B. subtilis*. Also in this case, it is probably due to the host organism being *E. coli*.

To explore potential factors contributing to this divergence, we initially considered differences in membrane lipid composition between *E. coli* and *B. subtilis*. However, lipid composition analysis revealed surprisingly similar compositions within the foci and septal areas of both bacteria, predominantly comprising PE and CL. This observation suggests that factors beyond lipid composition may influence the observed shift in foci localization.

Exploring potential theories behind the proteins influencing YqhY localization opens avenues for understanding the intricate mechanisms at play. YqhY lacks a transmembrane domain, prompting the question of how it engages with the membrane. Drawing parallels with the case of Asp23 discussed in section 1.4, it is plausible that YqhY binds to a transmembrane protein, and the differential distribution of this protein could explain YqhY's localization at the poles in *Bacillus* and the septal area in *E. coli*. Alternatively, YqhY may engage in a direct protein-protein interaction with ACC. In the case of *B. subtilis*, the proven localization of at least one of ACC's subunits, AccA, at the poles aligns with YqhY's localization in this region. For *E. coli*, if YqhY is able to interaction with its ACC, it is conceivable that ACC localizes elsewhere, possibly near the septal area, explaining the behavior of YqhY. Lastly, the prospect of a yet-to-be-identified third type of interaction adds an element of mystery to the intricate network of proteins influencing YqhY localization. Further exploration is crucial to unravel these intricate protein dynamics and their impact on cellular localization.

4.2. Overexpression of YqhY causes a shift in localization

The overexpression of YqhY in *E. coli*, aimed at mimicking the high expression levels observed in *B. subtilis*. Given the absence of YqhY in *E. coli*'s genetic makeup and its high expression in *B. subtilis*, we tried to replicate the physiological conditions observed in the latter bacterium by artificially inducing YqhY expression in *E. coli* using IPTG. This approach aimed to provide invaluable insights into the interplay between YqhY and its cellular environment, shedding light on its role and behavior in bacterial systems. However, our initial experimental endeavor yielded an unexpected outcome: while we observed an increase in fluorescence intensity, indicative of successful YqhY overexpression, we also noted a deviation from the anticipated localization pattern. This discrepancy prompted us to repeat the experiment and elucidate the observed shift in YqhY behavior over time.

The subsequent replication of the experiment, characterized by the concentration of high-intensity foci at the poles of bacterial cells, provided the anticipated results. In fact, overall the bacteria showed high resemblance with literature results [7], which underscored the consistency of YqhY's behavior. Furthermore, our quantitative analysis provided additional layers of validation, revealing a significant increase in the number of foci and a significant shift in their spatial distribution from the septal region to the poles.

The observed shift in the localization of foci from the septum to the poles in *E. coli* upon overexpression of YqhY underscores the need for further investigation into the underlying mechanisms governing this phenomenon. Given that *E. coli* with overexpressed YqhY exhibit a closer resemblance to the conditions observed in *B. subtilis*, it is prudent to prioritize these results over those obtained from uninduced colonies for subsequent studies. This shift in focus location suggests intricate regulatory mechanisms at play, which warrant thorough exploration to elucidate the functional implications of YqhY in bacterial physiology. Importantly, our findings highlight the consistent formation of foci by YqhY irrespective of induction status, indicating its propensity to organize into distinct cellular structures under various conditions. Furthermore, the replication of high-intensity foci formation at the poles in *E. coli* under conditions mimicking those in *B. subtilis* reinforces the notion of a conserved pattern of YqhY behavior across bacterial species, underscoring its potential significance in cellular processes.

Another aspect investigated was the impact of YqhY overexpression on bacterial growth rates. This study aimed to compare growth rates between cultures with induced YqhY and those without induction. The hypothesis, rooted in YqhY's role as an inhibitor of ACC in *B. subtilis*, suggested that overexpression would lead to decreased fatty acid synthesis and, consequently, slower growth. However, interpretation of the growth rate results proved inconclusive.
Given the inconclusive results obtained from the investigation into the impact of YqhY overexpression on bacterial growth rates in the current study, it would be prudent to consider conducting a parallel experiment in *B. subtilis*. The rationale behind this proposal lies in the potential variations in regulatory mechanisms and cellular processes between different bacterial species. This approach may reveal species-specific responses and shed light on whether YqhY's influence on bacterial growth rates is more relevant or pronounced in the context of *B. subtilis*.

4.3. Exposure to triclosan increases foci intensity

The experimental study focused on understanding the impact of the fatty acid synthesis inhibitors cerulenin and triclosan on the subcellular localization of YqhY. The objective was to impair fatty acid metabolism and examine how alterations in lipid biosynthesis resulting from the specific modes of action of these antibiotics might influence the behavior of YqhY.

In the investigation of growth with fatty acid synthesis inhibitors, the initial experiment yielded inconclusive results, challenging the hypothesis that inhibitors downstream of ACC would lead to more intense foci. The unexpected lack of foci in the negative control contradicted our previous findings, prompting a critical examination of potential factors influencing this behavior. The aging of the plate, being four weeks old, emerged as a plausible explanation for the atypical response observed in the negative control, as the bacteria might have adapted to the antibiotic, introducing an additional variable.

The experiment was then repeated by exposing the cells to triclosan through pads that were continuously monitored over time using a widefield microscope, capturing snapshots at 2.5-minute intervals throughout a 30-minute duration. This unveiled unexpected foci emergence in both the control and triclosan-exposed samples after the fourth time point. This indicated a significant shift approximately 10 minutes into the experiment and suggested that the initial lack of observable foci might have been attributed to the bacteria adapting to the pad environment, necessitating additional time for the manifestation of focal points.

As anticipated, the triclosan-exposed samples exhibited distinct foci with higher intensity compared to the control. The average foci intensity supported the hypothesis that, in the presence of the FabI inhibitor, YqhY increases its inhibitory activity on ACC, resulting in a higher intensity of the foci. The significant proportion of foci localized in the septal area further supported the observation of previous experiments.

To further elucidate the observed increase in foci intensity with triclosan exposure, additional experiments were designed. Chloramphenicol and mupirocin, inhibitors of protein translation, were employed to discern whether the elevation in intensity resulted from increased malonyl-CoA concentration or a general halt in cell growth. The rationale was based on the hypothesis that malonyl-CoA plays a pivotal role in influencing YqhY's inhibitory activity.

Results from the chloramphenicol and mupirocin experiments demonstrated low-intensity foci primarily at the septum, aligning with anticipated outcomes. This suggested that the initial increase in intensity might indeed be linked to a low concentration of malonyl-CoA rather than a cessation of cell growth. However, to confirm this hypothesis, it would be essential to measure the concentration of malonyl-CoA throughout all experimental conditions. The observed lower intensity for both antibiotics, around 1200 AU, potentially indicated a lower concentration of YqhY in the sample.

Interestingly, the decreasing number of foci over time may indicate a decrease in ACC inhibition. To support these findings, further investigations using *B. subtilis* are recommended, emphasizing the importance of cross-species validation. This approach ensures the consistency and applicability of observed patterns, enhancing the robustness and reliability of conclusions drawn from the experiment. In summary, while the initial experiment presented challenges, the subsequent refined conditions and additional experiments provided a clearer understanding of YqhY's dynamic response to fatty acid synthesis inhibitors.

4.4. Conclusion

From the obtained results, a compelling conclusion emerges, YqhY forms foci near the membrane, suggesting that YqhY holds the potential to function as a post-translational regulator in the fatty acid pathway of *B. subtilis.* The confirmed formation of foci by YqhY, especially under induction by triclosan, aligns with the speculated hypothesis that YqhY exerts its inhibitory activity on ACC when in the focal state. This implies a dynamic role for YqhY in modulating fatty acid synthesis through its spatial localization. Moreover, considering the parallels with the DUF322 family, it is conceivable that these proteins share common features, particularly the necessity for delocalization to trigger upregulation. In the case of YqhY, this upregulation manifests as an increase in ACC activity, while for Asp23, it leads to the upregulation of stress response genes. Given that *E. coli* served as the bacterial model in this study, it becomes imperative to validate and extend these findings by replicating the experiments in *B. subtilis*, the organism of interest. This additional step is crucial to establish the generalizability of the observed regulatory patterns and their significance in the specific context of *B. subtilis*.

5

Future prospects

The primary objective of this research was to unravel the intricate dynamics governing the role of YqhY in modulating ACC activity and its broader implications in balancing fatty acid synthesis within bacterial cells. The study focused on understanding the regulatory mechanisms, specifically exploring YqhY's responsiveness to membrane signals and its potential as an ACC inhibitor.

To extend and deepen the investigation, future research could include successfully visualizing YqhY in *B. subtilis*. Moreover, exposing *B. subtilis* to ACC inhibitors like triclosan and cerulenin, to see if the results match the ones obtained in *E. coli* would be imperative. Furthermore, understanding what would be the impact of malonyl-CoA depletion on YqhY localization, could shed light on its dynamic behavior under different stress-inducing conditions. Moreover, investigating the colocalization of ACC and YqhY could provide crucial information about their spatial interactions within the cellular environment, offering a comprehensive perspective on the regulatory network involved in fatty acid synthesis. Finally, diving into the potential abilities of YqhY to form filaments, could broaden our understanding of the mechanisms involved in the regulatory pathway. These proposed experiments aim to contribute to a more comprehensive understanding of the complex interplay between YqhY and ACC, opening avenues for further exploration in bacterial regulatory pathways.

5.1. Effects of fatty acid synthesis inhibitors in *B. subtilis*

For future prospects, it would be imperative to replicate the antibiotic experiments conducted in *E. coli* within the context of *B. subtilis*. This replication would serve to validate the findings obtained in *E. coli* and ensure their generalizability to *B. subtilis*. Employing a similar experimental protocol, with slight adjustments tailored to accommodate the unique growth requirements of *B. subtilis*, would be essential.

Likewise in this case, to clarify whether the increase in foci-intensity is due to an increase in the concentration of malonyl-CoA resulting from the inhibition of FabI or if it was a consequence of a general halt in cell growth, a control experiment using chloramphenicol and mupirocin should be conducted. *B. subtilis* containing our plasmid should be exposed to the two antibiotics through pads containing SMG medium and spectinomycin. Furthermore, it would be imperative to measure the concentration of malonyl-CoA at the different stages to ensure that there is a significant increase and decrease of the molecule.

5.2. Effects of malonyl-CoA depletion on YqhY localization

To substantiate our hypothesis positing that YqhY relocates to the cytoplasm in response to ACC upregulation, a targeted experiment is warranted. Specifically, we propose inducing malonyl-CoA depletion as a means to trigger increased ACC activity, thereby prompting YqhY relocation. This can be achieved by overexpressing FabD, an enzyme responsible for catalyzing the conversion of malonyl-CoA into malonyl-ACP. By enhancing the expression of FabD, we anticipate a reduction in malonyl-CoA levels, consequently leading to elevated ACC activity. Subsequent monitoring of YqhY localization under these conditions would provide valuable insights into its dynamic behavior in response to fluctuations in malonyl-CoA concentrations.

To accomplish this, we would construct a plasmid housing the *fabD* gene from *B. subtilis* along with a lacI inducer sequence. The lacI sequence serves to facilitate the overexpression of the plasmid as needed, induced by IPTG. Utilizing Gibson Assembly, we would integrate the *fabD* gene, obtained through PCR from the *B. subtilis* 168 genome using specific primers, into pDR111 [Supplementary Figure A.14]. Once the plasmid is assembled, it can be introduced into *E. coli* DH5α for transformation. Subsequently, the engineered plasmid can undergo further transformation into *B. subtilis* 168, which already harbors our pDR111_yqhY_GFP2_noAmyE construct. The detailed transformation protocol is outlined in the methods section.

Not to be overlooked is the fact that using overexpression of FabD in *B. subtilis* to deplete malonyl-CoA creates an interesting dynamic that may have a limiting influence on acyl carrier protein (ACP) in the context of the cell. It is possible that increasing FabD expression will have a disproportionate impact on malonyl-ACP formation, which could restrict the amount of ACP available for fatty acid synthesis's downstream stages. This may have a substantial impact on the balance and control of fatty acid synthesis as well as cellular lipid metabolism. In light of this, careful observation is needed to ensure that this is possible.

5.3. Co-localization and interaction between ACC and YqhY

Our hypothesis revolves around the potential direct or indirect interaction between YqhY and ACC, a notion supported by compelling evidence. Notably, deletion of YqhY has been shown to yield suppressor mutations in all four subunits of ACC, indicating a functional interplay between these proteins [7]. Furthermore, both YqhY and AccA, a key subunit of ACC, have been observed to localize in a foci pattern at the poles of *B. subtilis* cells, suggesting a spatial relationship between them [17]. To dive deeper into this intriguing aspect, two complementary experiments could be pursued. Firstly, using fluorescent microscopy could showcase the co-localization of these proteins, potentially by tagging YqhY with GFP and one of the ACC subunits with mCherry or YFP. Secondly, employing a bacterial two-hybrid system offers a means to elucidate whether a direct protein-protein interaction exists between YqhY and ACC, thereby providing further insights into their functional relationship. Through these investigative approaches, we aim to gain a comprehensive understanding of the dynamics and interactions underlying fatty acid synthesis regulation in bacterial cells.

5.3.1 Co-localization of ACC and YqhY using fluorescence microscopy

Fluorescence microscopy stands as an indispensable tool for unraveling the intricacies of protein co-localization. In the context of exploring the interaction between ACC and YqhY, careful consideration of the ACC subunit to be monitored is paramount. Among the ACC subunits, AccA emerges as the most promising candidate for observation, as AccA's localization at the poles of the cell has been established [Supplementary Figure A.4][17]. Furthermore, the combination of AccA and AccD constitutes the carboxyltransferase unit of ACC, responsible for the pivotal conversion of acetyl-CoA into malonyl-CoA [9]. Consequently, it becomes more logical to focus on monitoring this specific unit in order to gain insights into the critical step in fatty acid synthesis.

In pursuit of visualizing this interaction, two distinct methodologies can be employed. The first involves the design of a singular plasmid, a genetic construct containing the sequences for *yqhy*, GFP, *ACC* and yfp/mCherry. Leveraging pDR111_yqhY_GFP2_AmyE as the vector and employing Gibson assembly, the ACC_yfp/mCherry sequence would be inserted after the GFP sequence. Alternatively, a two-plasmid strategy can be implemented, where one plasmid contains the yqhy_GFP fusion, and the second accommodates the ACC_yfp/mCherry fusion. The first plasmid would be the plasmid used for previous experiment: pDR111_yqhY_GFP2_AmyE. On the other hand, construction of the second plasmid involves Gibson assembly, with pDR111_yqhY_GFP2_AmyE as the vector and the *Bacillus subtilis* genome serving as the DNA template. Once the two plasmids are ready, two subsequential transformations are required to ensure that both tagged proteins are transduced correctly.

Both methodologies aim to uncover the dynamic co-localization patterns between ACC and YqhY, offering valuable insights into their potential molecular interaction within the intricate cellular landscape.

5.3.2 Direct protein-protein interaction

If the results from fluorescent microscopy reveal a notable co-localization between ACC and YqhY, it would provide a compelling basis for diving deeper into their potential molecular interaction. To analyze the direct protein-protein interaction between ACC and YqhY, a Bacterial Adenylate Cyclase Two-Hybrid (BACTH) system stands as a promising investigative tool [41].

The BACTH system, relying on the reconstitution of *Bordetella pertussis* adenylate cyclase (CyaA), is a versatile tool designed for in vivo exploration of protein-protein interactions. In this system, the CyaA enzyme is split into two fragments, T25 and T18, which are individually fused to proteins of interest. When these fusion proteins interact, bringing the T25 and T18 fragments into close proximity, they reconstitute the adenylate cyclase activity of CyaA. This reconstitution leads to the synthesis of cyclic AMP (cAMP) from ATP. The generated cAMP activates the cAMP-dependent regulatory protein (CRP), initiating the expression of reporter genes such as lacZ. The measurable activity of these reporter genes serves as an indicator of successful protein-protein interactions 5.1. The BACTH system, by harnessing the enzymatic activity of CyaA, provides a valuable platform for investigating and validating protein interactions within a bacterial context, contributing to our understanding of cellular signaling pathways and molecular interactions.

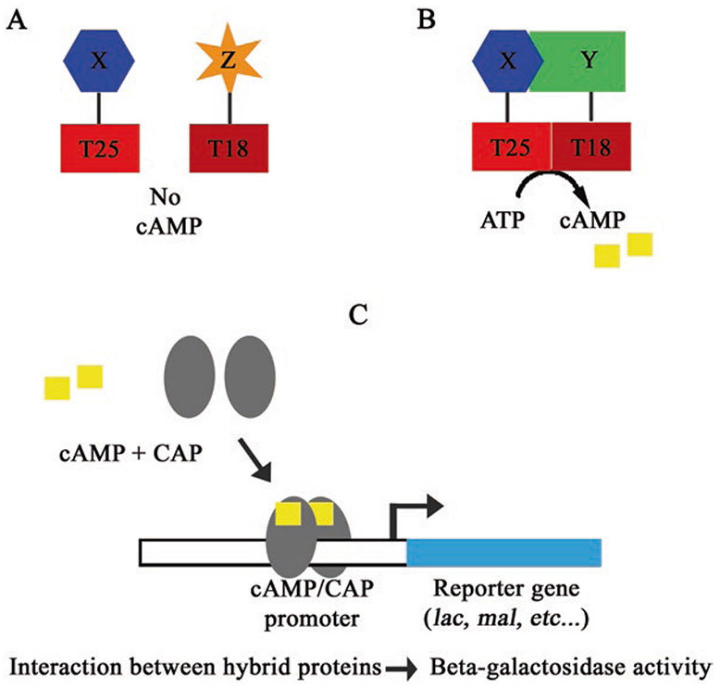

Figure 5.1: A schematic illustrating the principle of the BACTH assay. (a) Hybrid proteins that do not interact will not reconstitute adenylate cyclase activity. (b) Hybrid proteins that do interact will reconstitute adenylate cyclase activity, resulting in production of cAMP. (c) A depiction of cAMP binding CAP and positively regulating expression of a reporter gene [42].

To implement this process in our research we would have to individually amplify the coding sequences of AccA, AccB, AccC, AccD, and YqhY through PCR, using chromosomal DNA from *B. subtilis* as a template and specific primers. Subsequently, the PCR products would have to be subjected to PstI and BamHI digestion and cloned into the appropriate vectors, namely pUT18C, pUT18, pKT25, and pKNT25 [Supplementary Figure A.15]. This would result in the creation of either N-terminal (pKNT25 and pUT18) or C-terminal (pKT25 and pUT18C) fusions to the T25 or T18 fragment of the *B. pertussis* adenylate cyclase catalytic domain. We would use positive controls, featuring pUT18Czip and pKT25-zip, expressing T18 and T25 fused with the leucine zipper of GCN4, while negative controls utilizing the empty vectors pUT18C and pKT25. For co-expression, plasmids expressing either a T18 or a T25 construct for each protein would have to be co-transformed into cyaA-negative *E. coli*. For the BACTH system, three *E. coli* adenylate cyclase deficient reporter strains are commonly used in detecting protein–protein interactions: DHT1, DHM1, and BTH101 [42]. The transformed cells will be plated on selective media containing kanamycin, ampicillin, X-Gal, and IPTG, followed by incubation under specified conditions. Colonies displaying a blue coloration would be identified as indicative of positive protein-protein interactions, validating the BACTH system's utility in probing the molecular associations within the fatty acid synthesis pathway.

5.4. Investigating YqhY filamentation abilities

As we envision the future prospects for this research, diving into the intriguing possibility of YqhY filament formation emerges as an interesting avenue. The presence of the DUF322 domain in YqhY, a region known for its inherent self-interacting capabilities, suggests the potential for filamentation, a phenomenon observed in other members of the DUF322 family, such as Asp23 [20]. Filamentation, a dynamic process where proteins assemble into elongated structures, often plays a crucial role in regulating enzyme activity and cellular functions [43]. Thus, investigating whether YqhY also forms filaments presents a compelling question with far-reaching implications for understanding its role in bacterial physiology.

To initiate this exploration, a well-structured *in vitro* experiment could be conducted. Induction of YqhY expression using IPTG, followed by purification with a His-Tag, would enable the isolation of YqhY in a controlled environment [44]. Employing negative staining techniques would then provide an initial visualization of potential protein complexes, shedding light on the nature and organization of YqhY filaments. Subsequently, transmission electron microscopy could be employed to provide high-resolution images, enabling a detailed examination of the morphology and structural characteristics of YqhY filaments. This prospective investigation not only promises to unravel novel aspects of YqhY behavior but also contributes to the broader understanding of the functional significance of DUF322-containing proteins, potentially unveiling new strategies for modulating bacterial enzymes and cellular processes.

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[45] Karimova, G., Ullmann, A., and Ladant, D., “pKNT25.” AGER Lab, 2001. URL https://mbio.bas-net.by/cager/en/vectors-for-two-hybrid-systems/126-pknt25.html.

[46] Karimova, G., Ullmann, A., and Ladant, D., “pKT25.” AGER Lab, 2001. URL https://mbio.bas-net.by/cager/en/vectors-for-two-hybrid-systems/69-pkt25.html.

[47] Karimova, G., Ullmann, A., and Ladant, D., “pUT18.” AGER Lab, 2001. URL https://mbio.bas-net.by/cager/en/vectors-for-two-hybrid-systems/144-put18.html.

[48] Karimova, G., Ullmann, A., and Ladant, D., “pUT18C.” AGER Lab, 2001. URL https://mbio.bas-net.by/cager/en/vectors-for-two-hybrid-systems/48-put18c.html.

A

Appendix

A.1. Asp23 family protein sequence alignment

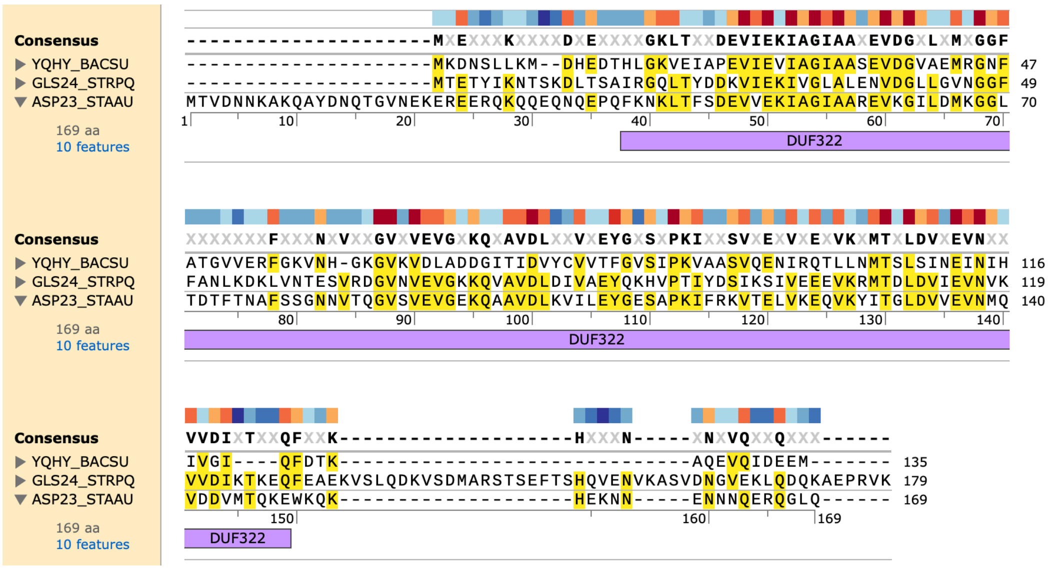

Figure A.1: Protein alignment of Asp23, Gls24 and YqhY, with visual representation of the DUF322 domain. Alignment done using Smith-Waterman algorithm in *SnapGene*.

A.2. DUF322 domain highly conserved in gram-positive bacteria

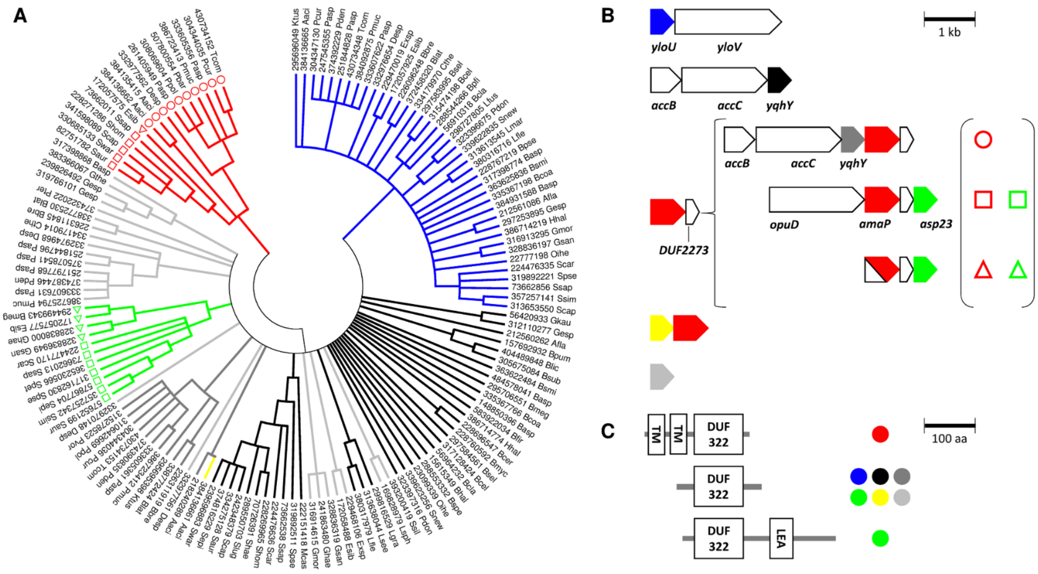

Figure A.2: Sequence analyses of the DUF322 protein domain for the order Bacillales. The bootstrap consensus tree (1000 replicates) for the DUF322 protein domain was inferred using the Neighbour-Joining method (A). Branches corresponding to partitions reproduced in less than 70% bootstrap replicates are collapsed. Taxon labels correspond to NCBI GI numbers and a four letter species abbreviation. Colouring of the branches is according to gene context in which DUF322 encoding genes are observed (see B). A schematic representation of conserved chromosomal contexts associated with DUF322 encoding genes is shown in (B). For the DUF322–DUF2273 gene cluster three distinct genetic contexts can be distinguished. To show the localization of the respective DUF2273 associated DUF322 domain proteins in the phylogenetic tree, symbols (open circles, open squares, and open triangles), shown in parenthesis to the right of the three different genetic contexts, are included as taxon markers in the tree presented in (A). Red symbols: AmaP-family; green symbols: Asp23-family. Although N-terminal transmembrane domains were always identified for the proteins encoded upstream of the DUF2273 encoding gene, a DUF322 domain was not always predicted with confidence as indicated by red-and-white colouring. Gene names are according to the S. aureus nomenclature: yloU, DUF322 domain protein; yloV, predicted dihydroxyacetone/glyceraldehyde kinase; accB, acetyl-CoA carboxylase subunit (biotin carboxyl carrier subunit); accC, acetyl-CoA carboxylase (biotin carboxylase subunit); yqhY, DUF322 domain protein; asp23, alkaline shock protein 23 and amaP, Asp23 membrane anchoring protein. The DUF2273 domain of unknown function represents a small c. 70 amino acids long protein that consists of two predicted transmembrane helices. Domain organization of DUF322 proteins within the Bacillales is shown in (C). Colours of the filled circles correspond to the colours used in (B) to correlate DUF322 protein domain organization with gene context conservation. TM: transmembrane domain; LEA: late embryogenesis abundant domain. [20]

A.3. Acetyl-CoA becomes limited in the absence of YqhY

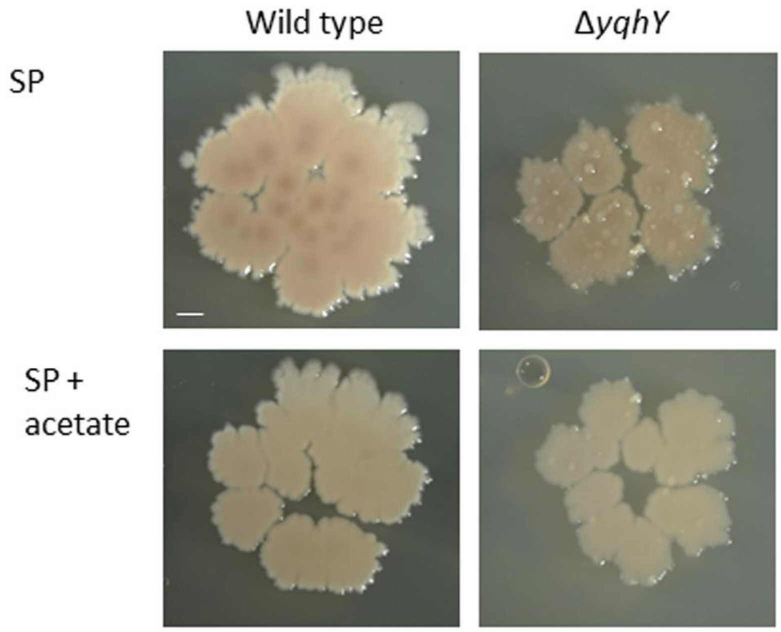

Figure A.3: From paper by Todter et el., effect of acetate on the yqhY mutant. The wild type strain 168 and the isogenic yqhY mutant GP1468 were cultivated on SP plates without or with added acetate (0.5%). Scale bar, 1 mm.

A.4. Acetyl-CoA Carboxylase subunit A localizes at the poles in *B. subtilis*

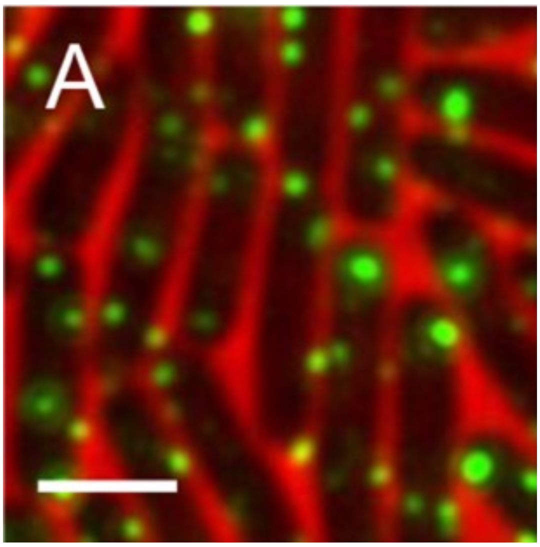

Figure A.4: A gene construct of the functional AccA-GFP expressed from the native accDA promoter was introduced into each of the wild-type [17]

A.5. Strains

Table A.1: Strains used in this research

Experiment	Strain	Description	Reference
Cloning	*E. coli* DH5α	Laboratory cloning strain	Greg Bokinsky lab
	E. coli DH5α + pBbE5k_yqhy_msGFP2	Cloning strain and the plasmid BbE5K with yqhy insert	This study
Microscopy	*E. coli* NCM 3722	Laboratory wild-type strain	Greg Bokinsky lab
	E. coli NCM 3722 + pBbE5k_yqhy_msGFP2	Wild-type strain and the plasmid BbE5K where msGFP2 is inserted to the C-terminus of the yqhy gene	This study
	B. subtilis 168 + pBbE5k_yqhy_msGFP2	Wild-type strain and the plasmid BbE5K where msGFP2 is inserted to the C-terminus of the yqhy gene	This study

A.6. Plasmid BbE5k_PlsB_msGFP2

The plasmid BbE5k_PlsB_msGFP2 used as backbone for this work. This plasmid is a work of the Greg Bokinsky lab.

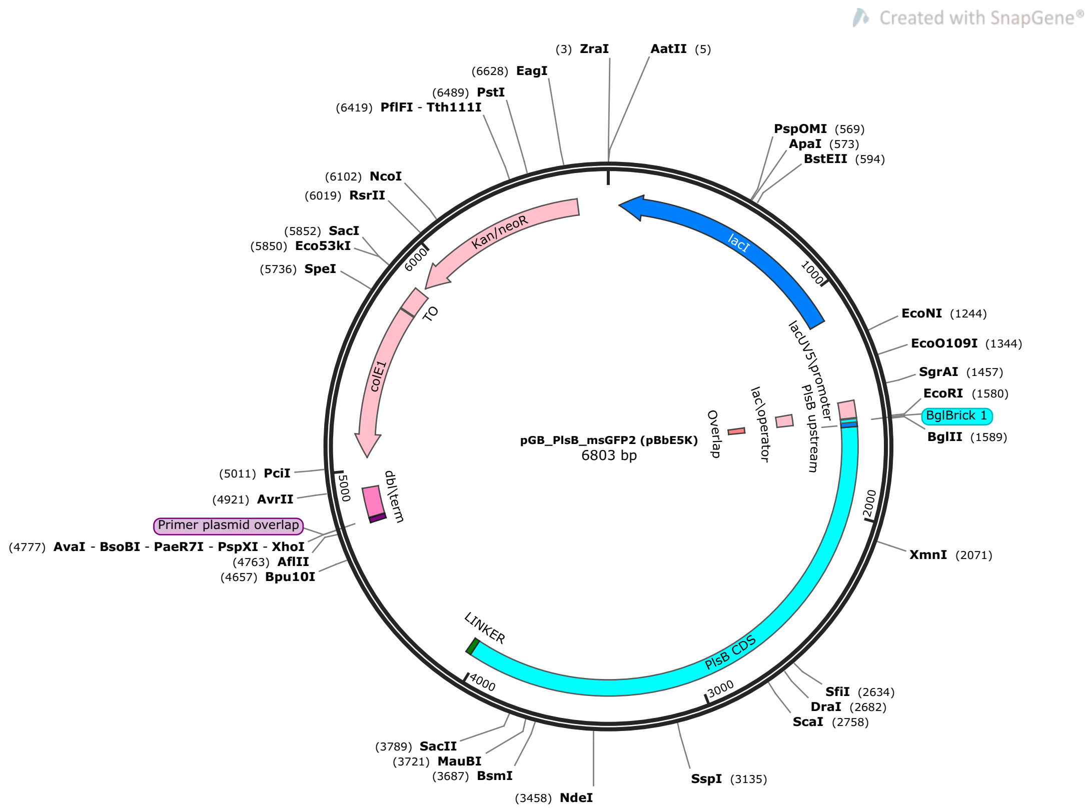

Figure A.5: Plasmid BbE5k_PlsB_msGFP2 with a start of origin (colE1), resistance for kanamycin (Kan/neoR) and an IPTG inducible site (lacI). This plasmid is the work of the Greg Bokinsky lab.

A.7. Matlab code for image analysis

Contents


```
clear all
```

Ask for what the experiment is being analyzed

```
prompt = {'What is the name of the experiment?'};
dlgtitle = 'Input';
dims = [1 36];
definput = {''};
answer = inputdlg(prompt,dlgtitle,dims,definput);

clear dims dlgtitle definput prompt %clearing variable not used further
```

Load images and define Phase-contrast and Fluorescent

```
[file,path] = uigetfile(".tif"); %prompt user for location of fluorscence image stack
Fullfile_F = fullfile(path,file);

Info = imfinfo(Fullfile_F); %finds the number of images in the stack
for n = 1:size(Info,1) %loop to load all the images
    F_image(:,:,n) = imread(Fullfile_F,n);
end

clear file path Fullfile_P Fullfile_F info %clearing variable not used further
```

Loading the Mesh that was created using MicrobeJ

Asking for the file that includes the mesh you generated with MicrobeJ

```
[file,path] = uigetfile(".mat"); %asks the user which file to upload
Mesh_data = load(fullfile(path,file)); %loads the actual mesh

% Creates new variables in the base workspace from those fields.
% (autogenerated by Matlab, does what I need)
vars = fieldnames(Mesh_data);
for i = 1:length(vars)
    assignin('base', vars{i}, Mesh_data.(vars{i}));
end

clear directory_path Directory file vars %clearing variable not used further
```

Foci Analysis

```
%final data array
Data = {'Image','Cell Number', 'Total cell Number','Number of Foci','Total Intensity of Foci','Number of pixels','Length','Width', 'Foci Location'}; %creating the header for easier reading of
Location = {'Image','Cell Number', 'Total cell Number','Number of Foci','Total Intensity of Foci','Number of pixels', 'Foci Location'}; %creating the header for easier reading of data: number

%Threshold parameters
Threshold_factor = 1.6;
Size_Threshold = 6;

%size of pixels in um/pixel units
prompt = {'Pixelsize:'};
dlgtitle = 'Input';
dims = [1 20];
definput = {'0.79'};
Output = inputdlg(prompt,dlgtitle,dims,definput);
Pixelsize = str2num(Output{1,1});

%preallocating arrays and setting variables
Backgrounds = zeros(1, size(Mesh_data.Experiment.Bacteria,1));Cytoplasm_intensity = zeros(size(Mesh_data.Experiment.Bacteria,1),25);iter = 2; iter_2 = 2;cell_count = 1; Masks = {};

Slices = zeros(1,size(Mesh_data.Experiment.Bacteria,1)); Labels = {};
for n = 1:size(Mesh_data.Experiment.Bacteria,1)
    Slices(1,n) = Mesh_data.Experiment.Bacteria(n).POSITION.slice; %Collects all the slices into one cell array
    Labels{1,n} = Mesh_data.Experiment.Bacteria(n).IMAGE.label;
end

cnt_unique = hist(Slices,unique(Slices)); %Counts how often each label (slice) occurs
unique_a = unique(Labels, 'stable'); %finds each unique (string) label

Counter = 1;
for n = 1:size(F_image,3) %loop makes a index array for the bacteria (e.g. bacteria 1 - 32 are image 1, bacteria 33 - 86 are image 2 etc.)
    Cell_count(n,1) = Counter; %start at one
    Counter = Counter + cnt_unique(1,n) -1;
    Cell_count(n,2) = Counter;
    Counter = Counter + 1;
end

Debug = false ; %toggle debug mode, either "true" or "false"

for n = 1:size(F_image,3) %cycles through the images
    Temp_tot_mask = zeros(size(F_image,1), size(F_image,2));
    Temp_background = F_image(:,:,n);
    for i = Cell_count(n,1):Cell_count(n,2) %cycles through the by MicrobeJ identified cells for background and masks
        clear Temp_Mesh %needed because not all meshes are the same size, so simply overwriting doesn't work
        for j = 1:size(Mesh_data.Experiment.Bacteria(i).Contour,1)
            Temp_Mesh(j,1) = Mesh_data.Experiment.Bacteria(i).Contour(j).COORD.x / Pixelsize; %Coordinates from microbeJ are in um, therefore you need to divide by the pixelsize to convert the
            Temp_Mesh(j,2) = Mesh_data.Experiment.Bacteria(i).Contour(j).COORD.y  / Pixelsize;
        end

        Temp = poly2mask(Temp_Mesh(:,1), Temp_Mesh(:,2), size(F_image, 1),size(F_image, 2)); %Uses the cell mesh to make a mask
        Masks{n,i} = Temp; %stores the mask
        Bounding_box(i,:) = [min(Temp_Mesh(:,1)), max(Temp_Mesh(:,1)), min(Temp_Mesh(:,2)), max(Temp_Mesh(:,2))];

        if (min(Bounding_box(i,:)) < 0.51) && (min(Bounding_box(i,:)) > 0) %Values lower than 0.5 get rounded down which causes issues since 0 is not a valid coordinate
            Index = find(Bounding_box(i,:) == min(Bounding_box(i,:))); %Finds the lowest value
            Bounding_box(i,Index) = ceil(Bounding_box(i,Index)); %rounds up the lowest value
        elseif (min(Bounding_box(i,:)) <= 0)
            Index = find(Bounding_box(i,:) <= 0); %Finds the zero or negative value
            Bounding_box(i,Index) = 1; %sets it to 1
        end
        Temp_tot_mask(Temp == 1) = 1;
        Temp_background(Temp == 1) = 0;

    end

    Backgrounds(1,n) = median(Temp_background, 'all'); %background calculated from all non-cell pixels (aka non-zero pixels)
    Temp_Cell = F_image(:,:,n); %loads the fluorescent image
    Temp_Cell_subt_back = Temp_Cell - Backgrounds(1,n); %subtracts the background
```

Figure A.6: Greg Bokinsky Lab pipeline for fluorescence images. Quantitative approach to analyze the presence of foci in bacteria. Part 1.

```matlab
for i = Cell_count(n,1):Cell_count(n,2) %cycles through cells for foci analysis
    Temp_current_cell = Temp_Cell_subt_back;
    Temp_mask = Masks{n,i}; %loads the current mask
    Temp_current_cell(Temp_mask == 0) = 0; %sets all non-cell pixels to zero

    Cell = Temp_current_cell(Bounding_box(i,3):Bounding_box(i,4), Bounding_box(i,1):Bounding_box(i,2));

    %Get the width and length of the cell as determined by MicrobeJ
    Length = Mesh_data.Experiment.Bacteria(i).SHAPE.length;
    Width = Mesh_data.Experiment.Bacteria(i).SHAPE.width.mean;

    %finds median value, taken to be cytoplasm value (placeholder)
    Cell_d = double(Cell); %converts to double because NaN values are not allowed in uint16
    Cell_d(Cell_d == 0) = NaN; %Replaces zero with NaN, needed for nanmedian
    Cytoplasm_intensity(n,i) = median(Cell_d,'all','omitnan'); %median on all non-NaN(= non-zero) values

    Threshold = Cytoplasm_intensity(n,i)* Threshold_factor; %Calculates threshold

    Bin_Cell = imbinarize(Cell, (Threshold/65535)); %Binarizes image based on threshold
    Bin_Cell_size = bwareaopen(Bin_Cell, Size_Threshold,4);
    Clusters = bwconncomp(Bin_Cell_size); %Determines connectivity


    %If not a single pixel was above the threshold, the code will
    %assume the median value was too high and try a lower one
    if sum(Bin_Cell, 'all') == 0
        Low_threshold = Cytoplasm_intensity(n,i) * 0.8; %lower threshold value to find all the higher values that might be foci
        Bin_Cell = imbinarize(Cell, (Low_threshold/65535)); %Binarizes image based on threshold
        %The non-foci parts of the cell are found and the mean is
        %found
        Cytoplasm = Cell_d;
        Cytoplasm(Bin_Cell == 1) = NaN;
        Cyto_bin = imbinarize(Cytoplasm, 0);
        Cyto_av = Cell_d;
        Cyto_av(Cyto_bin == 0) = NaN;
        Average = mean(Cyto_av, 'all', 'omitnan');
        %This mean is used as a new cytoplasmic value to find foci
        New_threshold = Average * Threshold_factor;
        Bin_Cell = imbinarize(Cell, (New_threshold/65535));
        Bin_Cell_size = bwareaopen(Bin_Cell, Size_Threshold,4);
        Clusters = bwconncomp(Bin_Cell_size);
    end
    Number_foci = Clusters.NumObjects; %takes number of objects as number of foci

    % Determine foci location
    stats = struct2table(regionprops(Bin_Cell_size, 'Centroid'));

    if Number_foci > 0
        for f = 1:Number_foci
        Foci_centroid = stats.Centroid(f,:);

        Foci_pixels_separate = Cell;
        Foci_pixels_separate(Bin_Cell_size == 0) = 0; %Find all pixel values of the foci

        int = struct2table(regionprops(Bin_Cell_size, Foci_pixels_separate,'MeanIntensity'));

        Intensity_foci = int.MeanIntensity(f,:);
        % Define a threshold for determining pole vs. septal area (adjust as needed)
    Threshold = 0.3;

    % Calculate the radius of the cell
    Cell_area = (Bounding_box(i,2) - Bounding_box(i,1)) * (Bounding_box(i,4) - Bounding_box(i,3));
    Cell_radius = Length/2;

    % Calculate the properties of each connected component
    gray_matrix = mat2gray(Cell);
    binary_Cell = imbinarize(gray_matrix);
    props = regionprops(binary_Cell, 'Centroid');

    % Extract the centroid of the first cell
    Cell_centroid = props.Centroid;

    % Print some debug information
    disp(['Foci centroid: ', num2str(Foci_centroid)]);
    disp(['Cell area: ', num2str(Cell_area)]);
    disp(['Cell centroid: ', num2str(Cell_centroid)]);

    % Calculate the distance between the two points
    distance = sqrt((Cell_centroid(1)-Foci_centroid(1)).^2+((Cell_centroid(2)-Foci_centroid(2)).^2));

    % Print additional debug information
    % Print the distances matrix
    disp('distance between the Foci_centroid and Cell_centroid:');
    disp(distance);

    disp(['Cell radius: ', num2str(Cell_radius)]);

    disp(['threshold: ', num2str(Cell_radius * Threshold)]);
```

Figure A.7: Greg Bokinsky Lab pipeline for fluorescence images. Quantitative approach to analyze the presence of foci in bacteria. Part 2.

```
        for l = 1:size(Foci_centroid, 1)
            if any(distance(l) < Cell_radius * Threshold)
                FociLocation = 'Septal Area';

            else
                FociLocation = 'Pole';
            end
        end

        Location{iter_2,1} = n; %creating the header for easier reading of data
        Location{iter_2,2} = i; %number given to cell by Oufti
        Location{iter_2,3} = cell_count; %overall number of cell
        Location{iter_2,4} = Number_foci; %number of foci
        Location{iter_2,5} = Intensity_foci; %total intensity of all foci
        Location{iter_2,6} = FociLocation; % foci location within the cell

        iter_2 = iter_2 + 1;
        disp(['Foci Location: ', (FociLocation)]);
            end
        else
                FociLocation = 'None';

        end

        Foci_pixels = Cell;
        Foci_pixels(Bin_Cell_size == 0) = 0; %Find all pixel values of the foci
        Intensity = sum(Foci_pixels,'all'); %sums all pixel values for total intensity

        Pixels = sum(Bin_Cell_size, 'all'); %finds the number of pixels

        Data{iter,1} = n; %creating the header for easier reading of data
        Data{iter,2} = i; %number given to cell by Oufti
        Data{iter,3} = cell_count; %overall number of cell
        Data{iter,4} = Number_foci; %number of foci
        Data{iter,5} = Intensity; %total intensity of all foci
        Data{iter,6} = Pixels; %total number of pixels in all foci
        Data{iter,7} = Length; %length of the cell
        Data{iter,8} = Width; %mean width of the cell
        Data{iter,9} = FociLocation; % foci location within the cell

        if Debug == true %debug option if you want to check the foci masks
            %montage({Bin_Cell, Bin_Cell_size, adapthisteq(Cell,'clipLimit',0.06,'Distribution','uniform'),labeloverlay(adapthisteq(Cell,'clipLimit',0.06,'Distribution','uniform'),Bin_Cell_si
            figure;
            imshow(labeloverlay(adapthisteq(Cell,'clipLimit',0.06,'Distribution','uniform'),Bin_Cell_size,'Colormap',[0,1,0]), 'Border', 'tight')
            % Plot the centroids
            hold on; % Make sure we can add to the current axes
            plot(Foci_centroid(1), Foci_centroid(2), 'bo', 'MarkerSize', 10, 'MarkerFaceColor', 'b'); % Blue cross for foci centroid
            plot(Cell_centroid(1), Cell_centroid(2), 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r'); % Red circle for cell centroid
            hold off; % Stop adding to the current axes

            pause(4); %gives you time to look at the images
        end

        iter = iter + 1;
        cell_count = cell_count + 1;
    end

end

%clearing variables not used further
clear iter Debug Clusters Bin_Cell Bin_Cell_size Number_foci Cell Threshold Temp_current_cell Temp_Cell_subt_back Temp_background Temp_background Temp Temp_Mesh Temp_masks G i j n New_threshol
clear Output dims dlgtitle prompt definput
```

export data as Excel

```
excel_name = append('Image_Analysis_', answer{1,1},'.xlsx'); %makes a standard filename for your convenience

answer_save = questdlg('Do you want to save your data?','Export Data');
if isequal(answer_save,'Yes')
    [filename, pathname] = uiputfile(excel_name);
    fullname = fullfile(pathname, filename);
    % xlswrite(fullname, Data);
    writecell(Data,fullname);
else
    return
end

clear filename pathname fullname answer_save %clearing variable not used further

excel_name = append('Image_Analysis_Location_', answer{1,1},'.xlsx'); %makes a standard filename for your convenience

answer_save = questdlg('Do you want to save your data?','Export Data');
if isequal(answer_save,'Yes')
    [filename, pathname] = uiputfile(excel_name);
    fullname = fullfile(pathname, filename);
    % xlswrite(fullname, Data);
    writecell(Location,fullname);
else
    return
end
```

Figure A.8: Greg Bokinsky Lab pipeline for fluorescence images. Quantitative approach to analyze the presence of foci in bacteria. Part 3.

A.8. Plasmid DR111

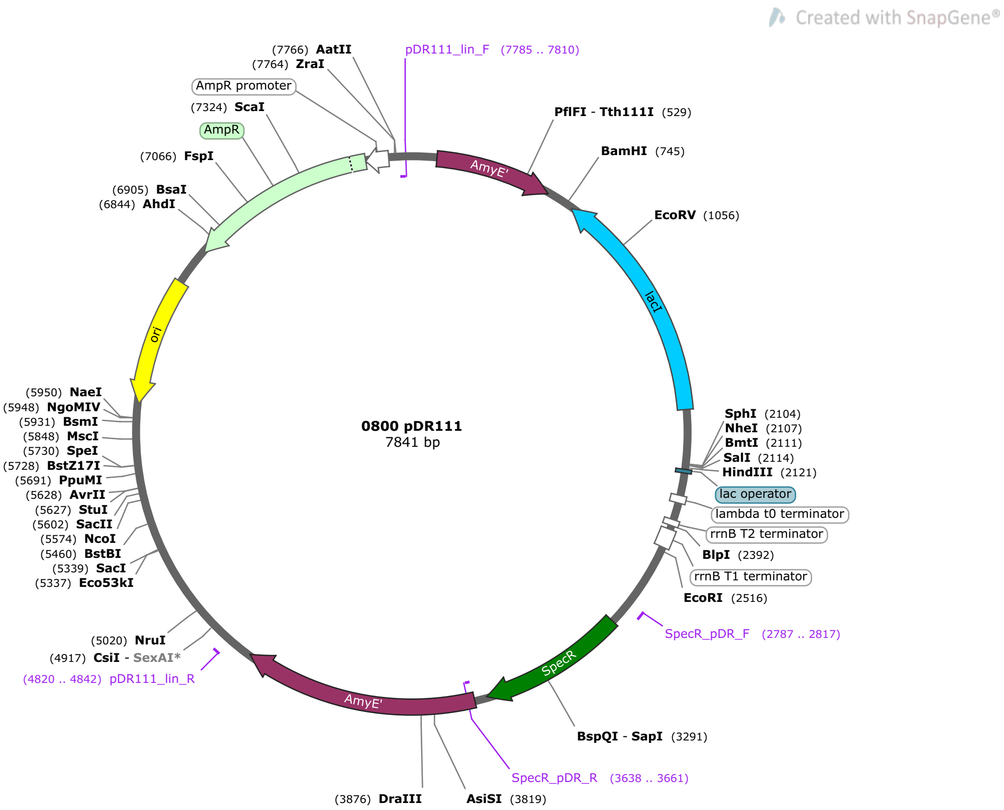

Figure A.9: pDR111 map created with *SnapGene*. It presents an origin (ori), resistance to both Ampicillin (AmpR) for *E. coli* and Spectinomycin (SpecR) for *B. subtilis*, alpha-amylase gene (AmyE) for genome integration and an IPTG inducible site (lacI).

A.9. *Bacillus subtilis* AmyE locus Map

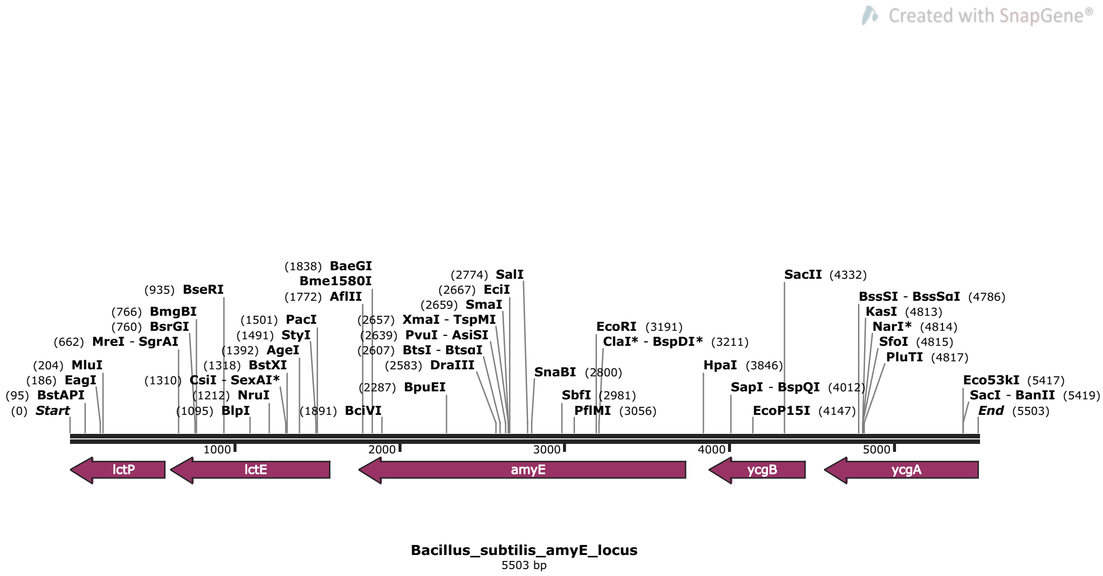

Figure A.10: *Bacillus subtilis* AmyE locus Map

A.10. First YqhY overexpression results

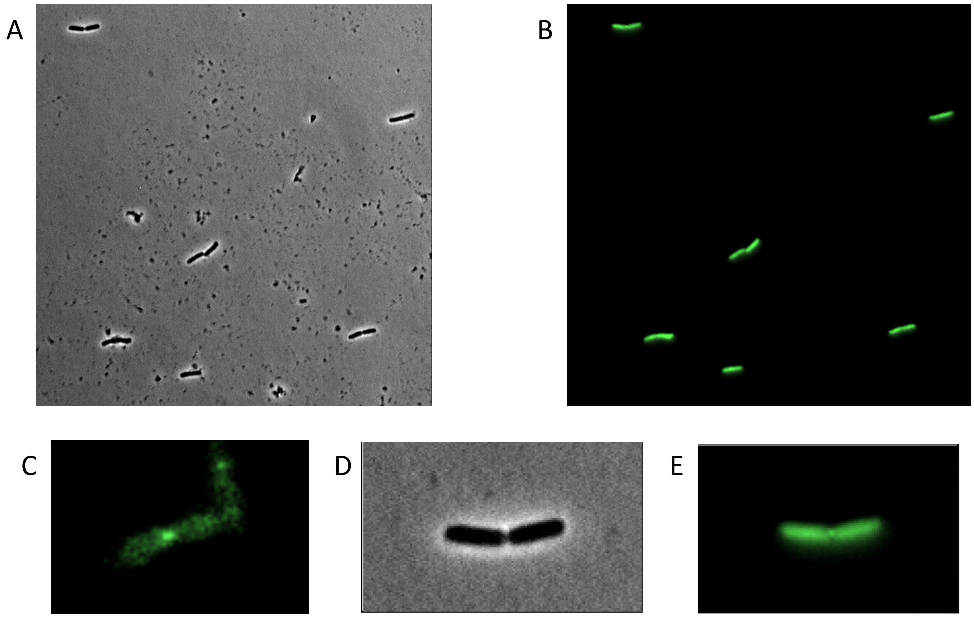

Figure A.11: Homogeneous distribution of yqhY_GFP2 after overexpression induced in IPTG in NCM3722. A. Phase contrast image with 100x objective. B. Fluorescence image with 100x objective. C. Magnification 30x of yqhY_GFP2 from section 3.1. D. Magnification 30x of Phase contrast image. E. Magnification 30x of fluorescence image.

A.11. Uninduced and induced YqhY foci intensities

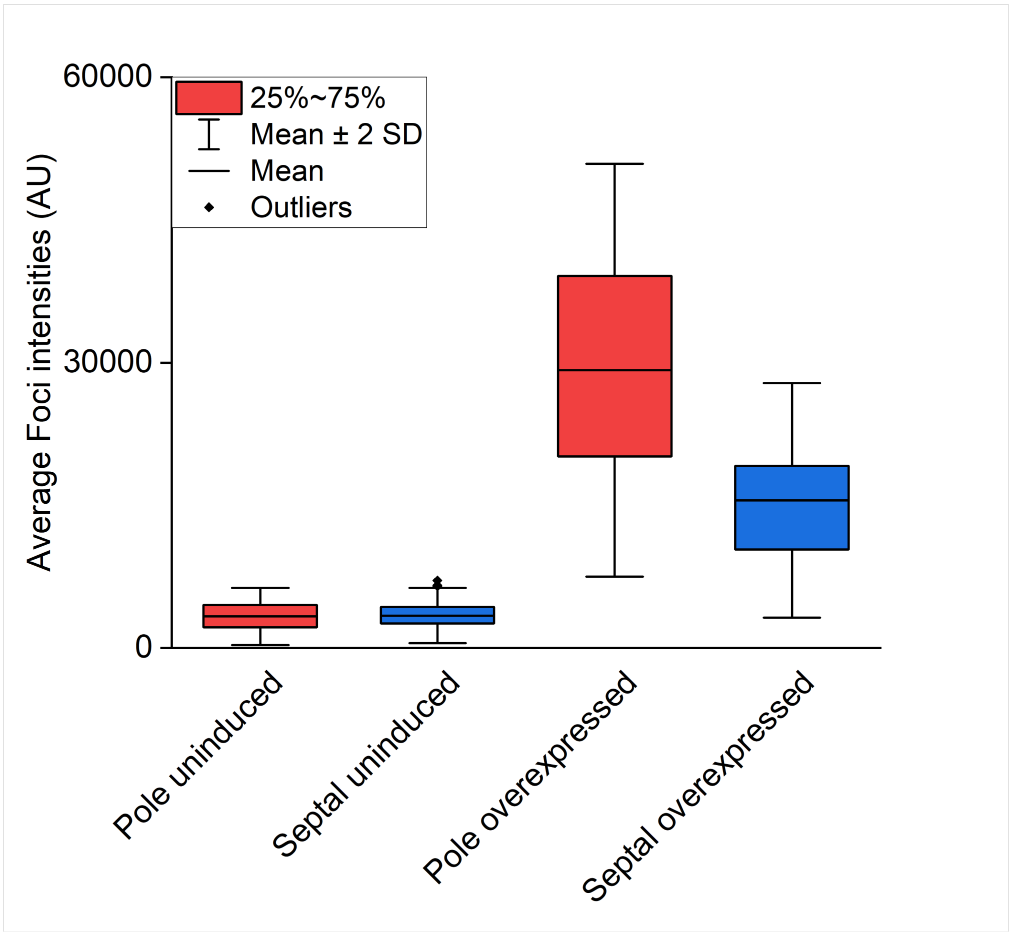

Figure A.12: Comparison between foci intensity at the septal area and poles in induced and uninduced YqhY results

A.12. Differences in number of foci according to different antibiotics

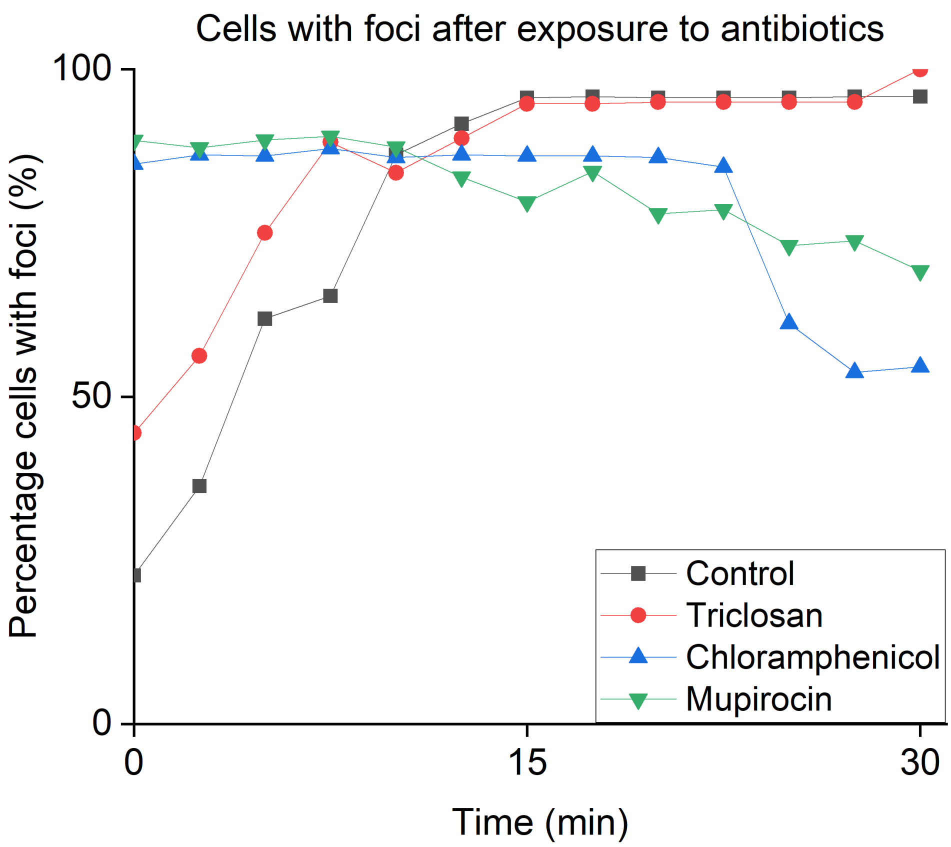

Figure A.13: According to the different antibiotics the number of foci decreases. As visible in the image, the number of foci increases in the cased of the control and triclosan, but it decreases as mupirocin and chloramphenicol are introduced in the system.

A.13. Plasmid DR111 with FabD insertion for malonyl-CoA depletion experiment

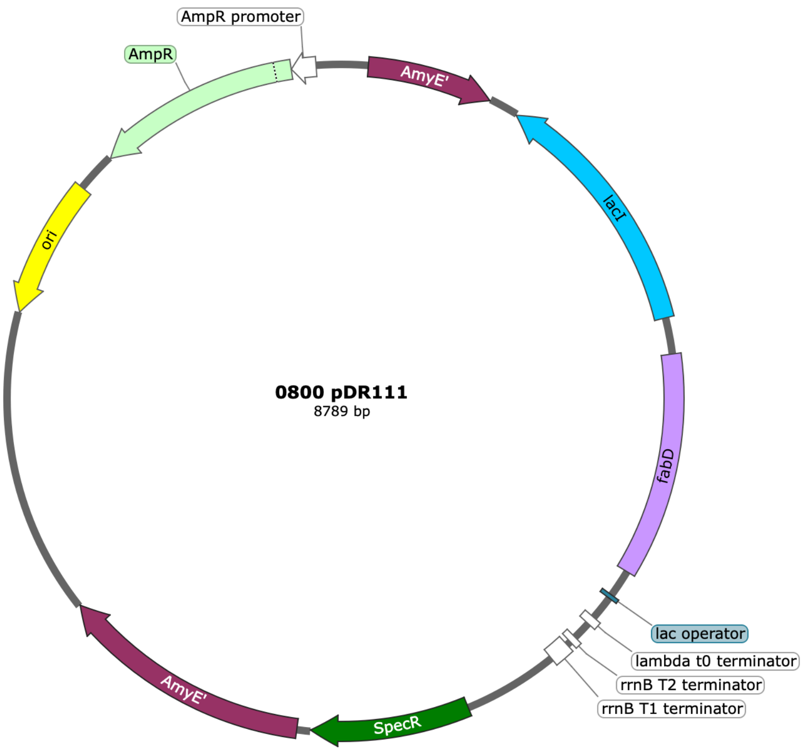

Figure A.14: pDR111_fabD map created with *SnapGene*. It presents an origin (ori), resistance to both Ampicillin (AmpR) for *E. coli* and Spectinomycin (SpecR) for *B. subtilis*, alpha-amylase gene (AmyE) for genome integration, *B. subtilis* 168 fabD and an IPTG inducible site (lacI).

A.14. Plasmids for BACTH

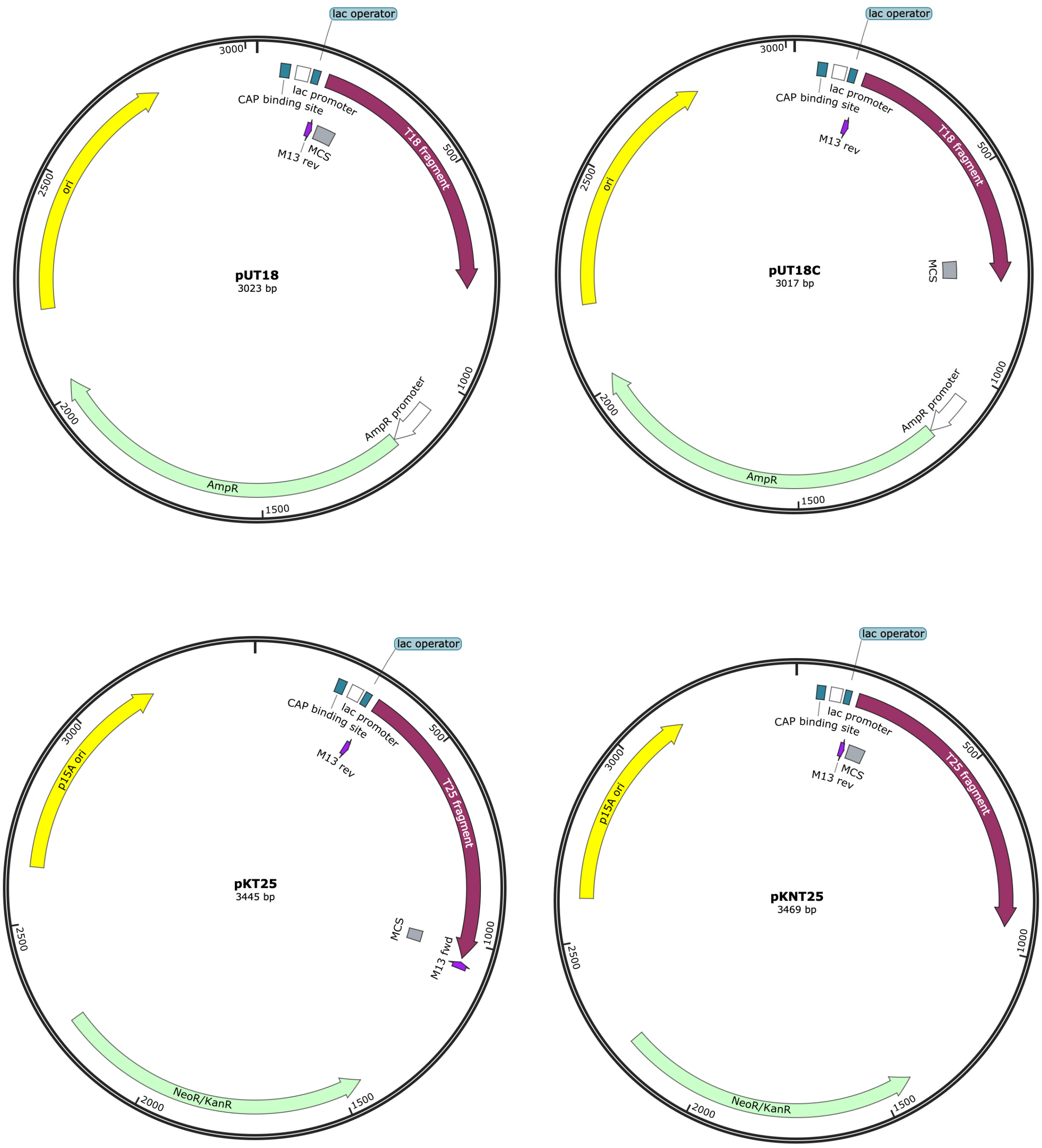

Figure A.15: Plasmids used for Bacterial two-hybrid experiment [41][45][46][47][48].